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Ayranci et al.

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(54) **WIDEBAND COUPLED INPUT IMPEDANCE
MATCHING LNA ARCHITECTURE**

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patent is extended or adjusted under 35
U.S.C. 154(b) by 692 days.

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H03F 1/56 (2006.01)
H03F 3/195 (2006.01)
H03F 3/72 (2006.01)

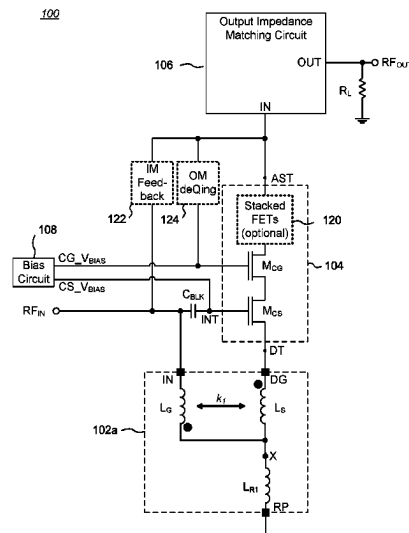
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(2013.01); **H03F 3/72** (2013.01); **H03F**
2200/294 (2013.01); **H03F 2200/451**
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CPC H03F 1/22; H03F 9/00
USPC 330/311, 310, 165, 276
See application file for complete search history.

(57) **ABSTRACT**

Circuits and methods for a radio frequency amplifier, such as
an LNA, that include a wideband coupled input impedance
matching network. One embodiment includes a first inductor
coupled between a first terminal and a first node, the first
terminal couplable to a degeneration terminal of an amplifier
core; a second inductor coupled between a second terminal
and either the first node or a second node, the second
terminal couplable to an input terminal of the amplifier core;
a third inductor coupled between the first node and a third
terminal, the third terminal couplable to a reference poten-
tial; and, in a variant embodiment, a fourth inductor coupled
between the second node and a fourth terminal, the fourth
terminal couplable to the reference potential; wherein the
first inductor and the second inductor are mutually coupled.
Some embodiments allow multiple modes to allow tradeoffs
of gain versus linearity and NF characteristics.

10 Claims, 19 Drawing Sheets



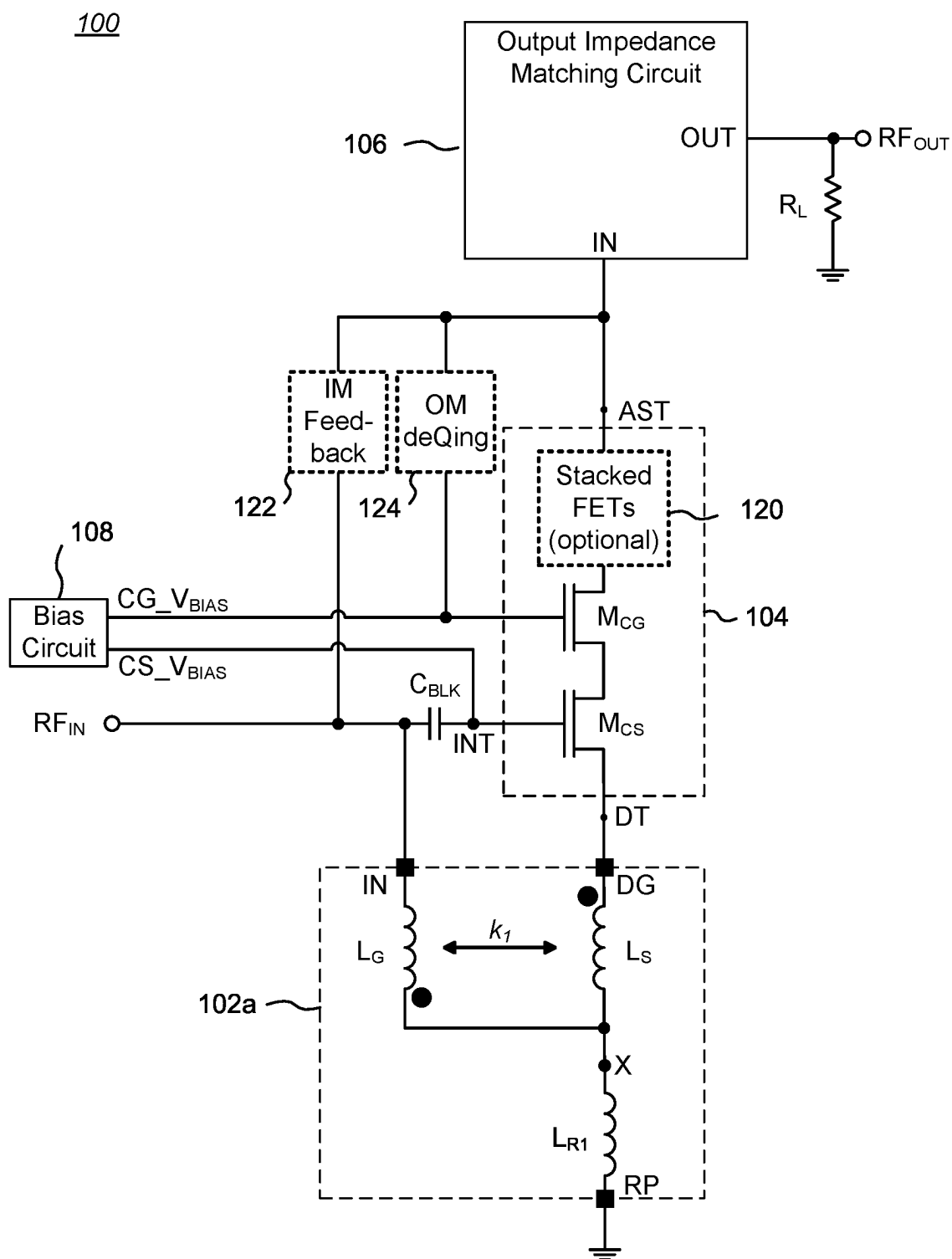


FIG. 1

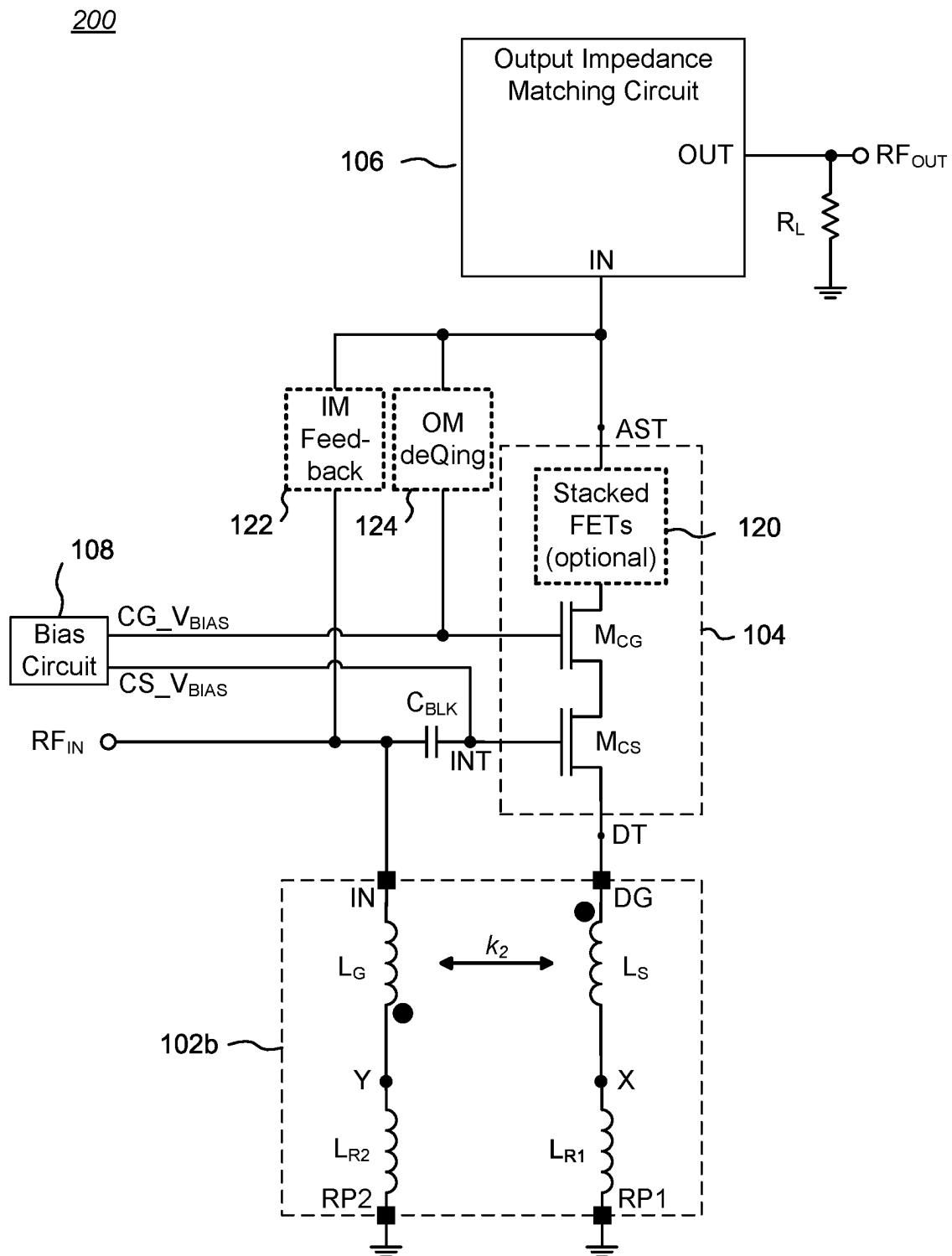


FIG. 2

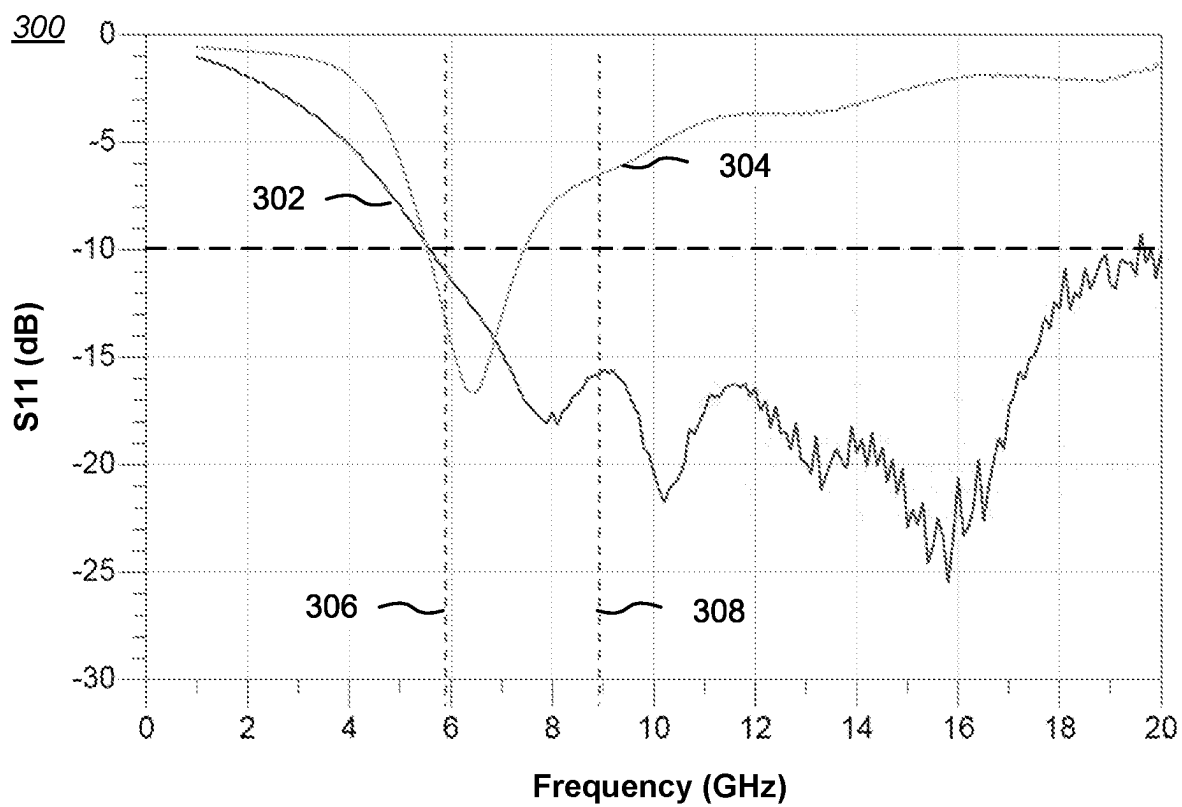


FIG. 3A

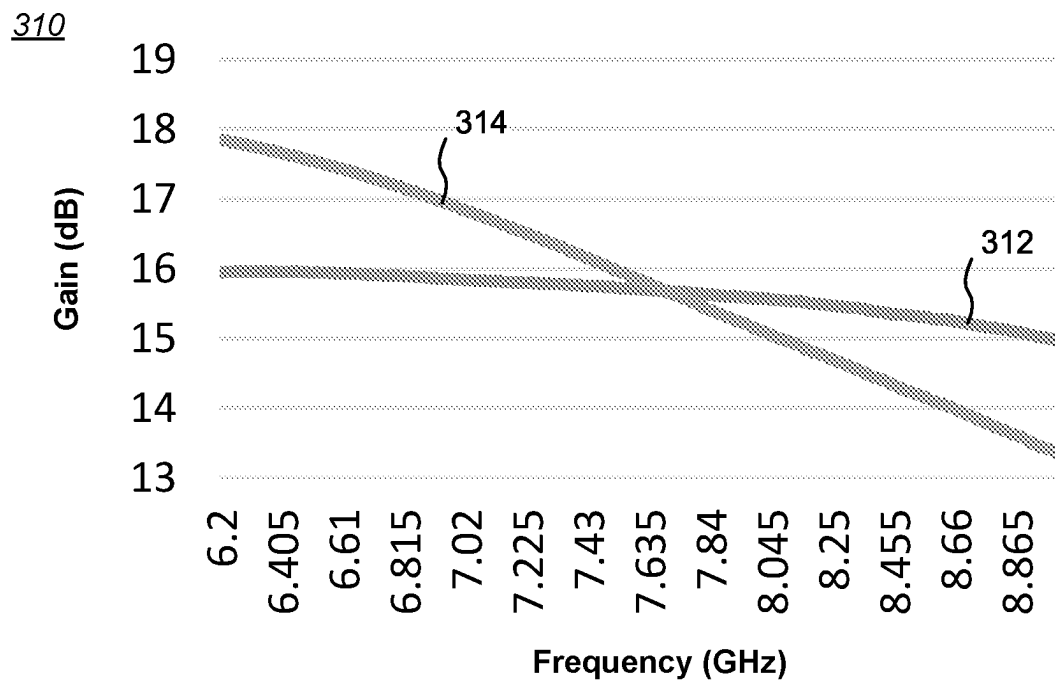


FIG. 3B

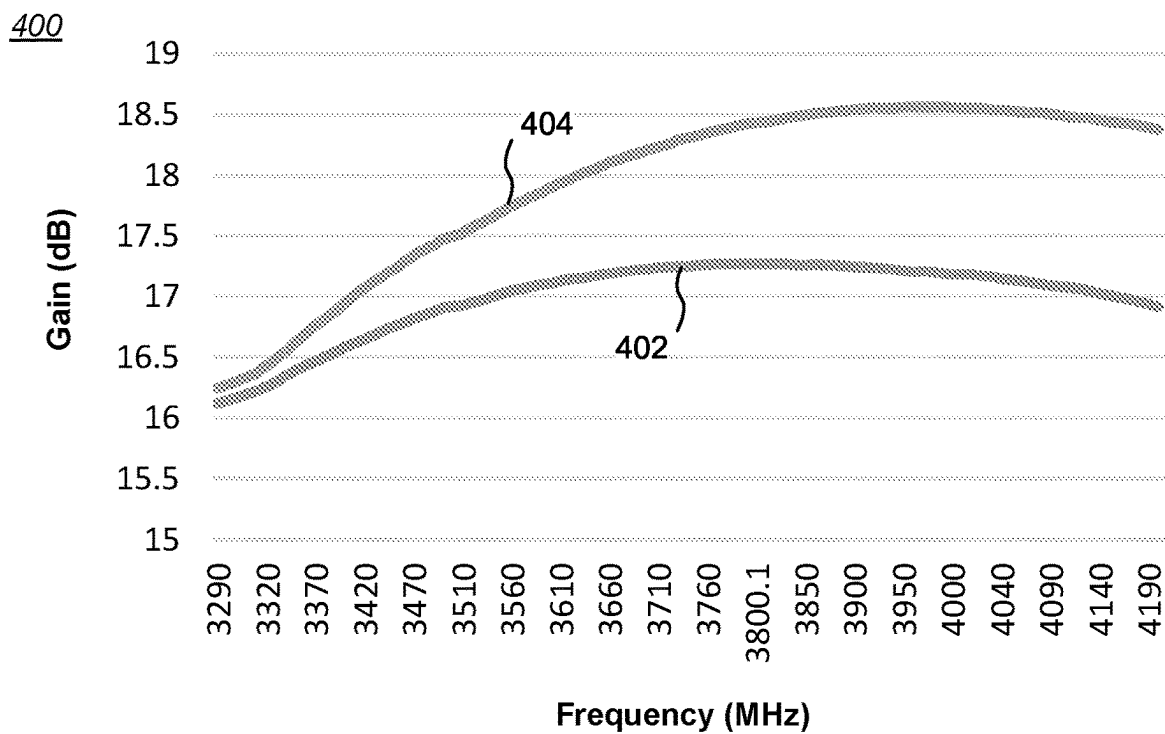


FIG. 4A

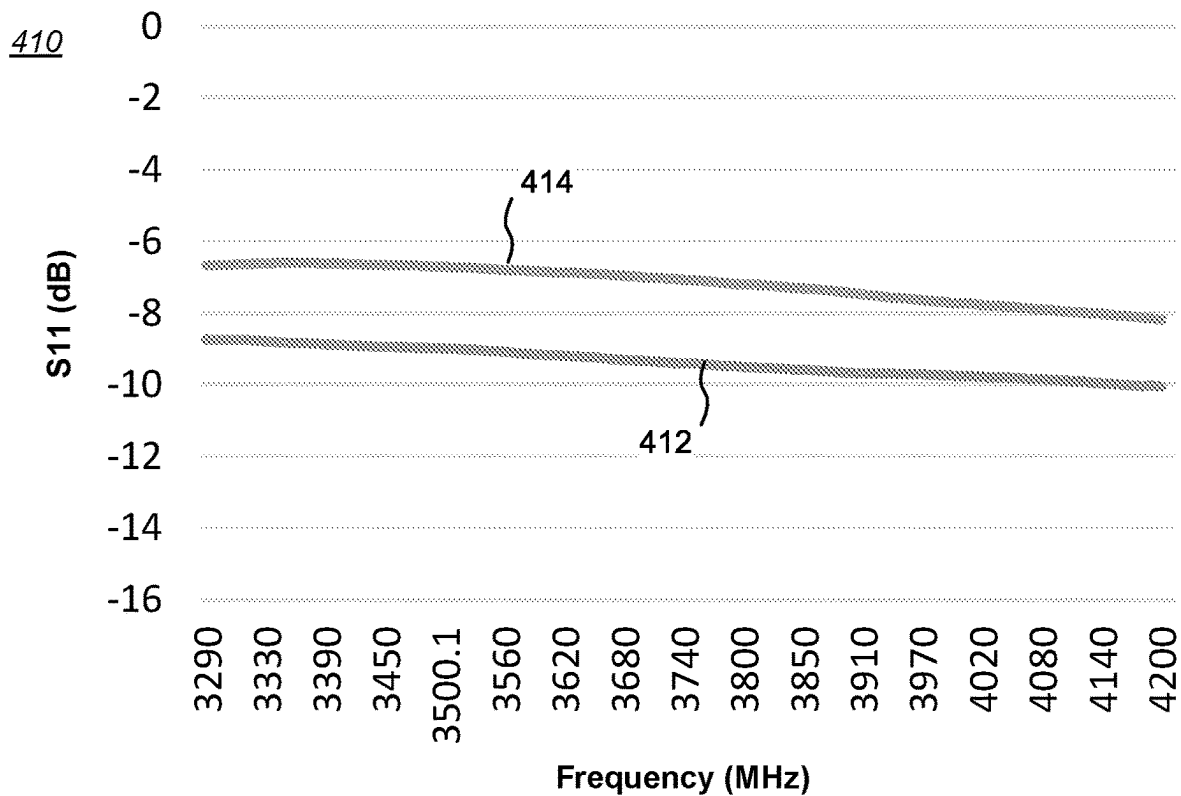


FIG. 4B

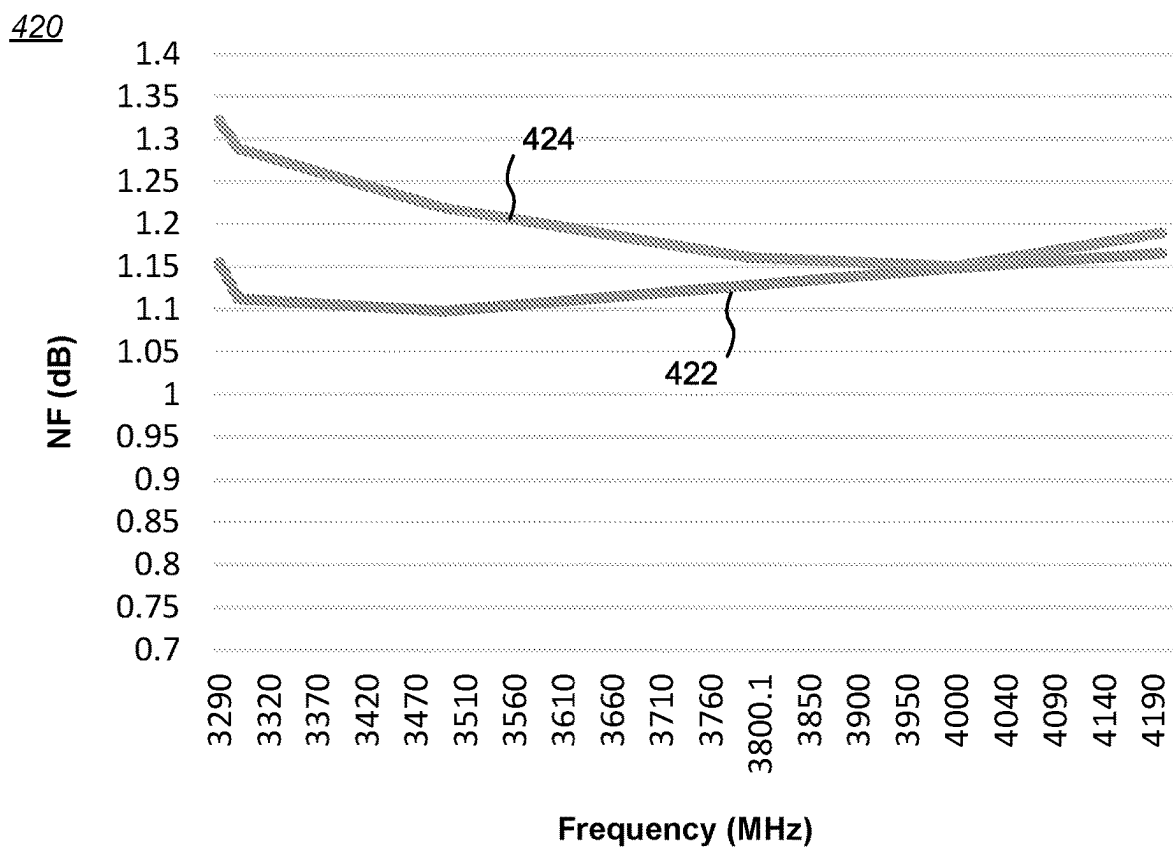


FIG. 4C

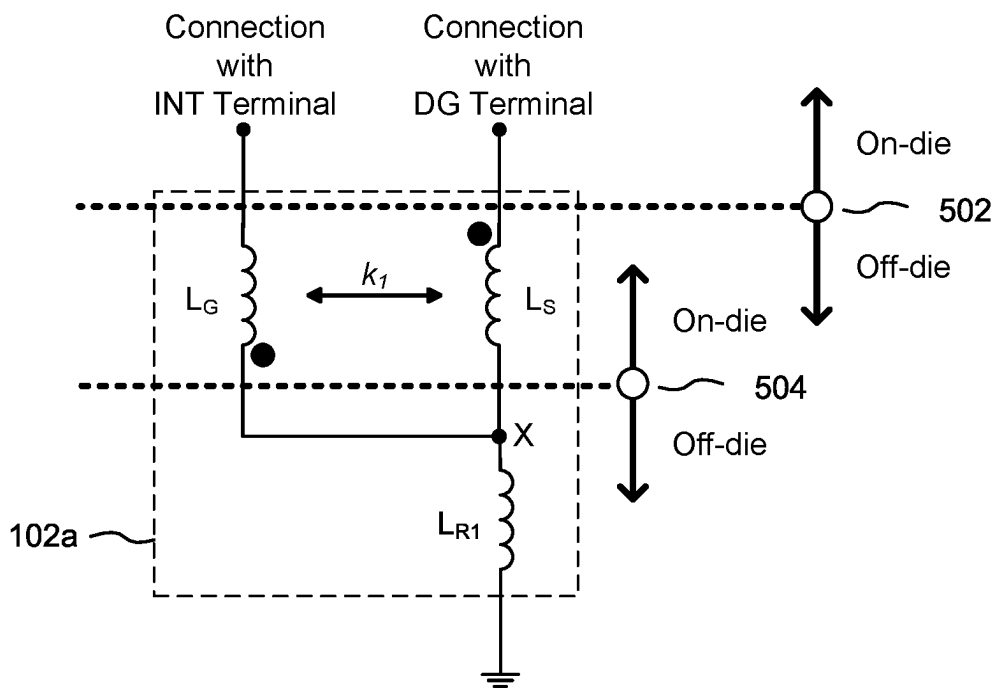


FIG. 5A

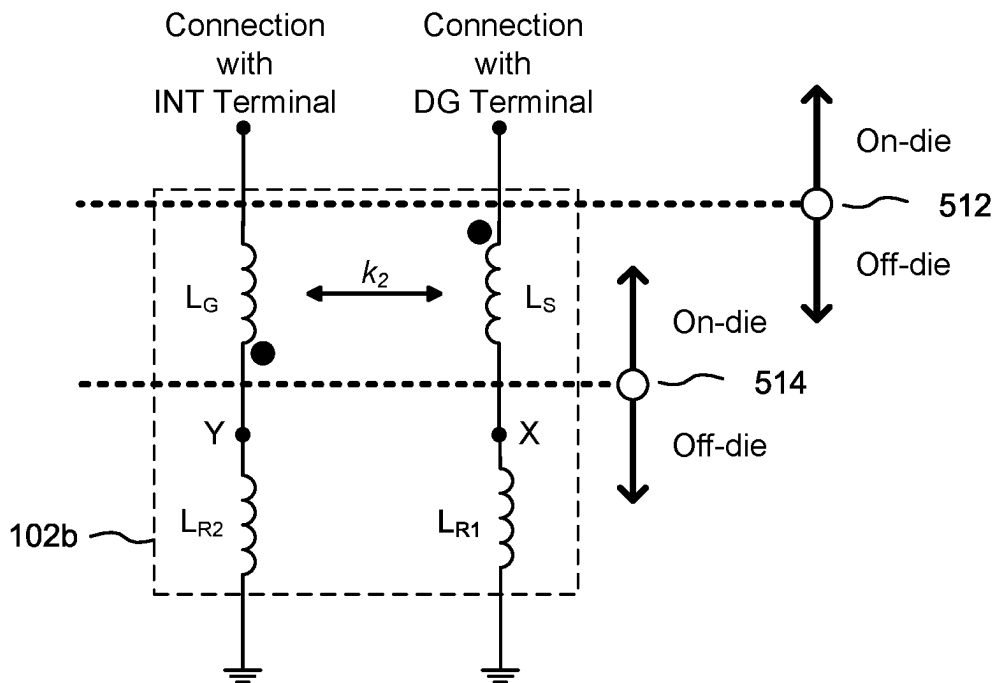


FIG. 5B

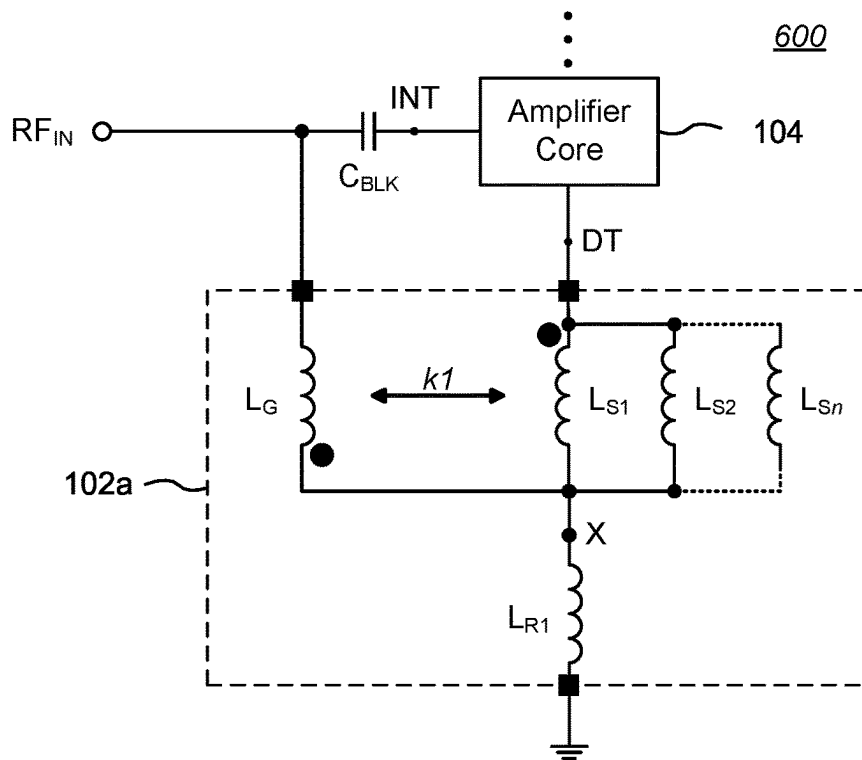


FIG. 6A

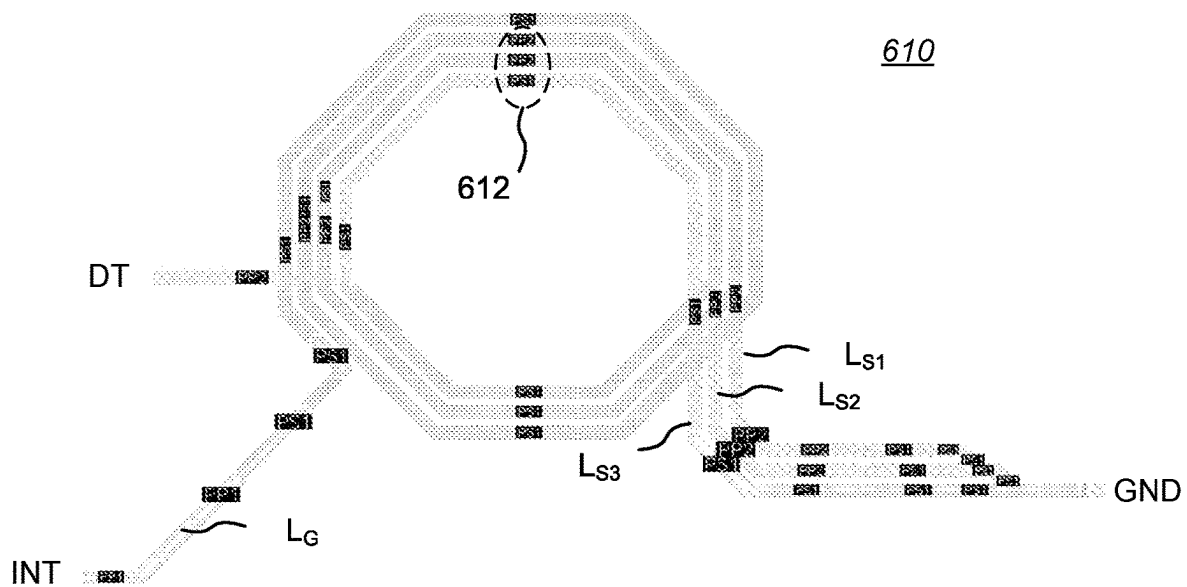


FIG. 6B

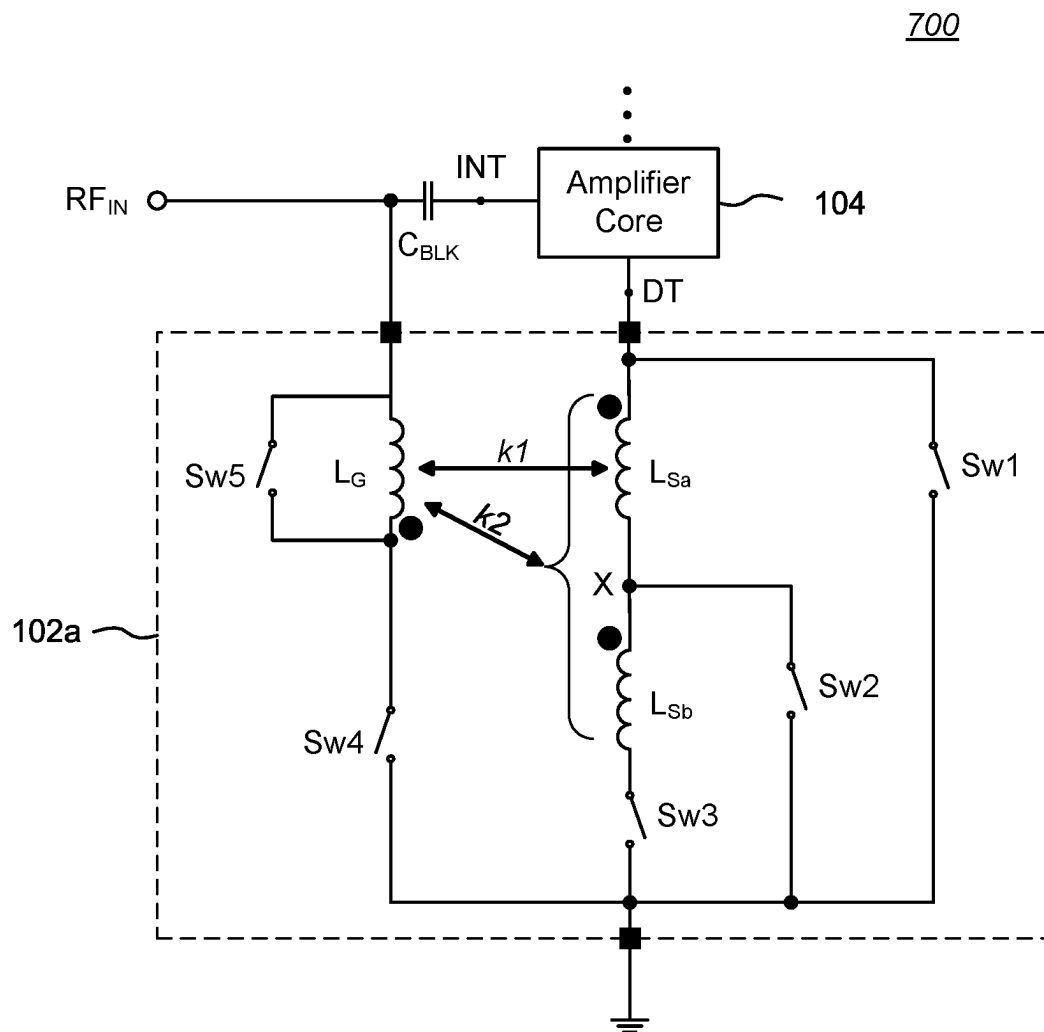


FIG. 7

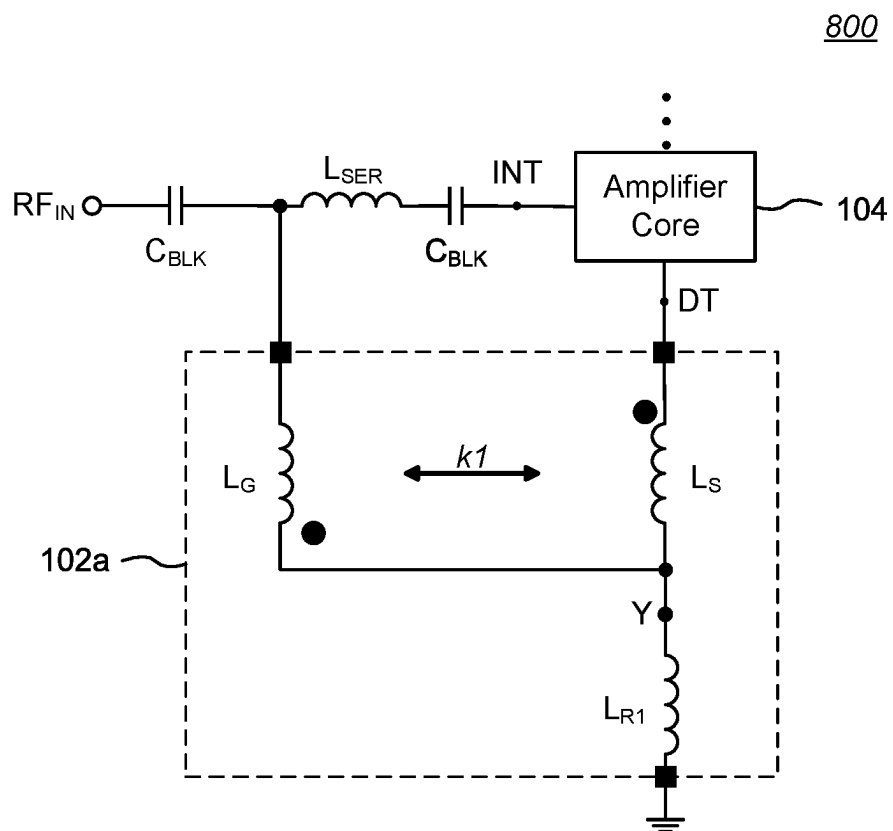
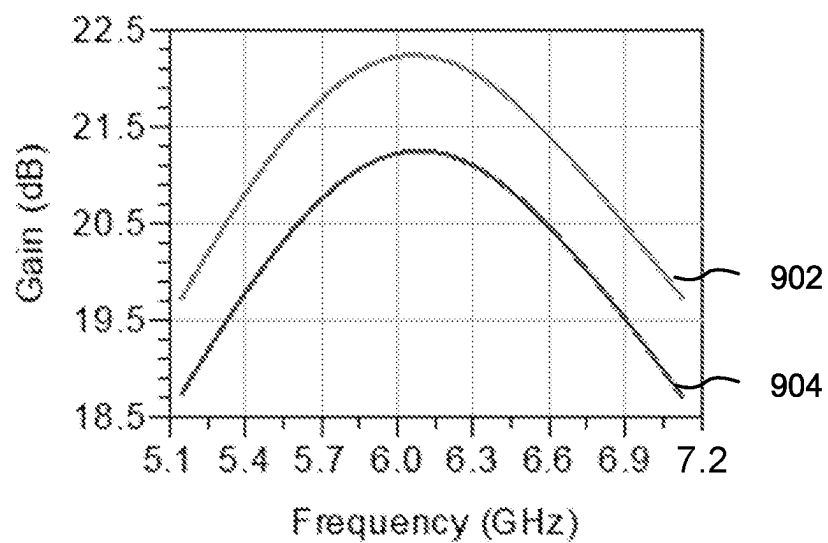
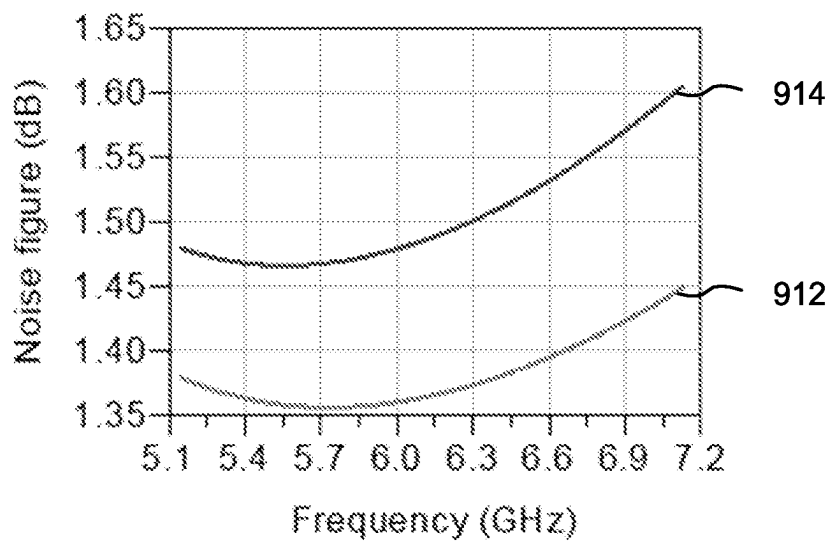


FIG. 8

900**FIG. 9A****FIG. 9B**

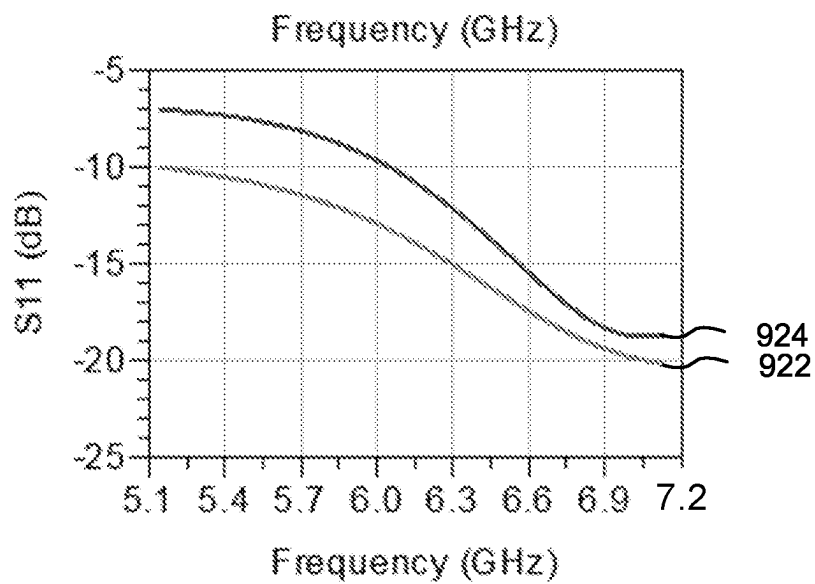
920

FIG. 9C

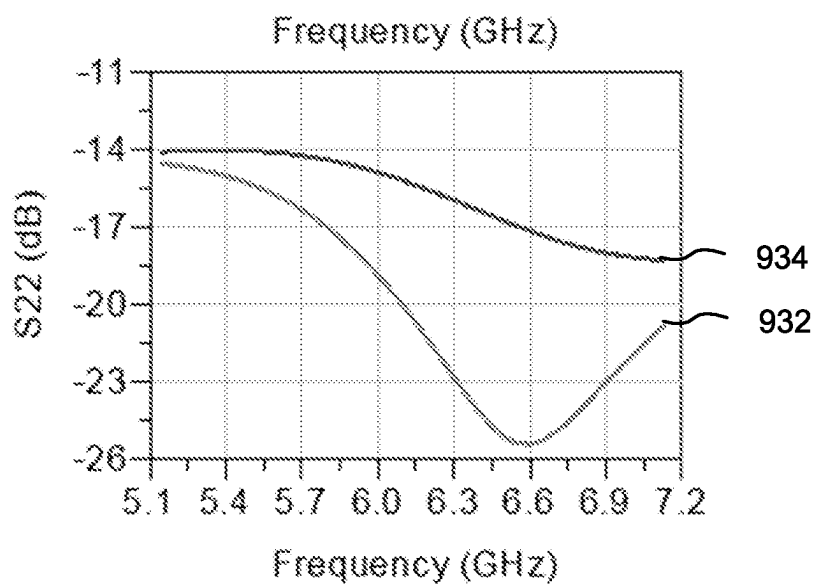
930

FIG. 9D

940

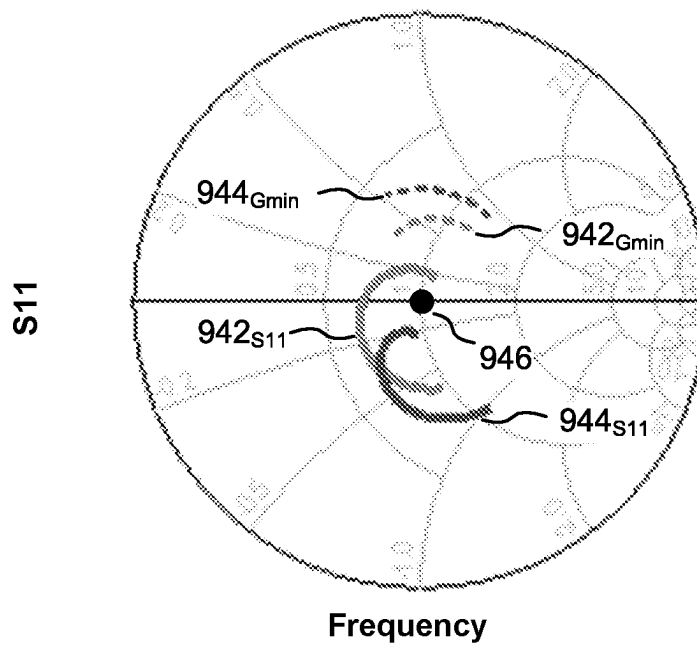


FIG. 9E

950

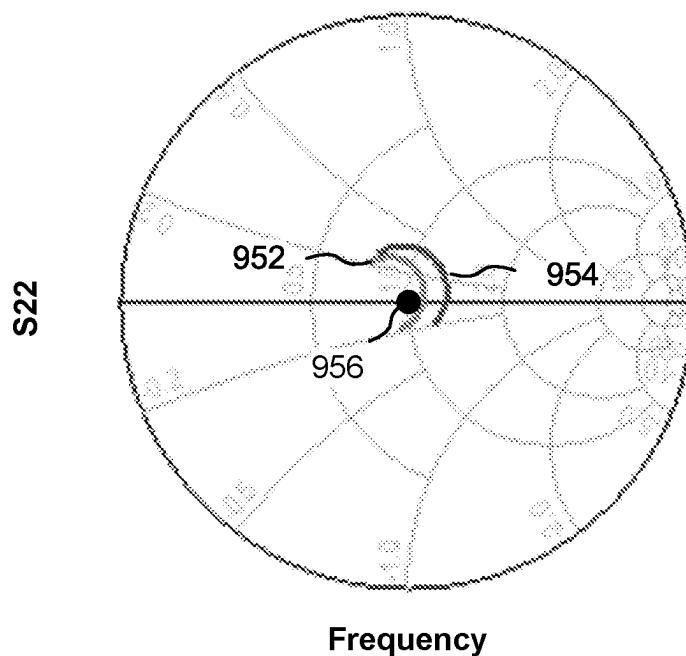


FIG. 9F

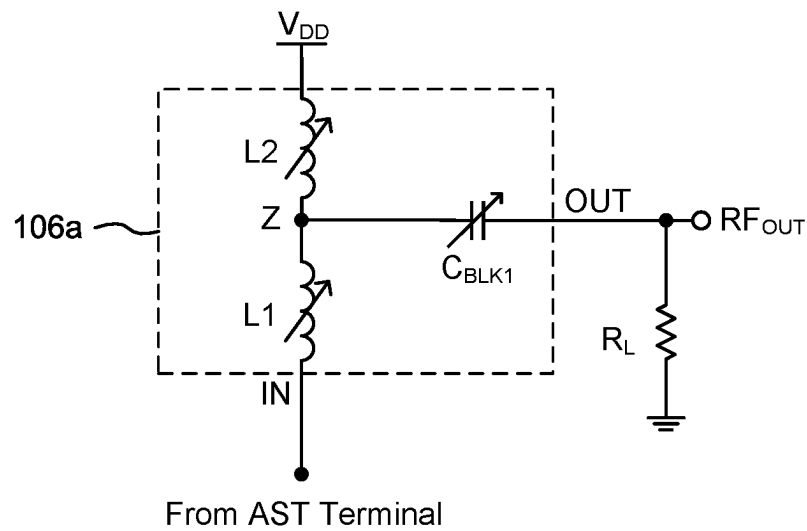


FIG. 10A

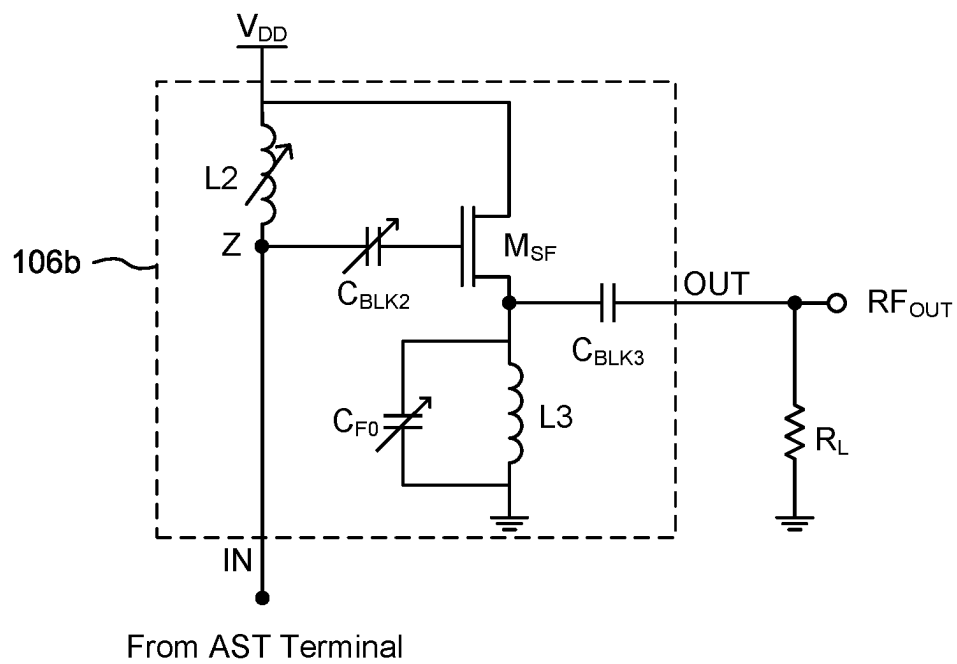


FIG. 10B

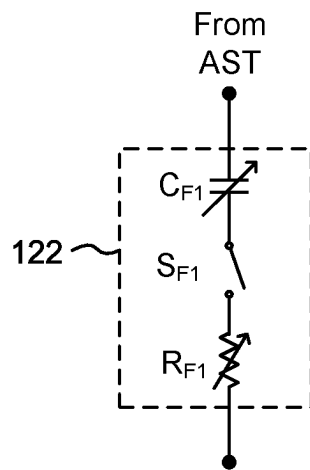


FIG. 11

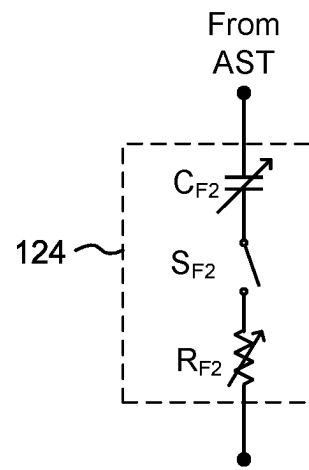


FIG. 12

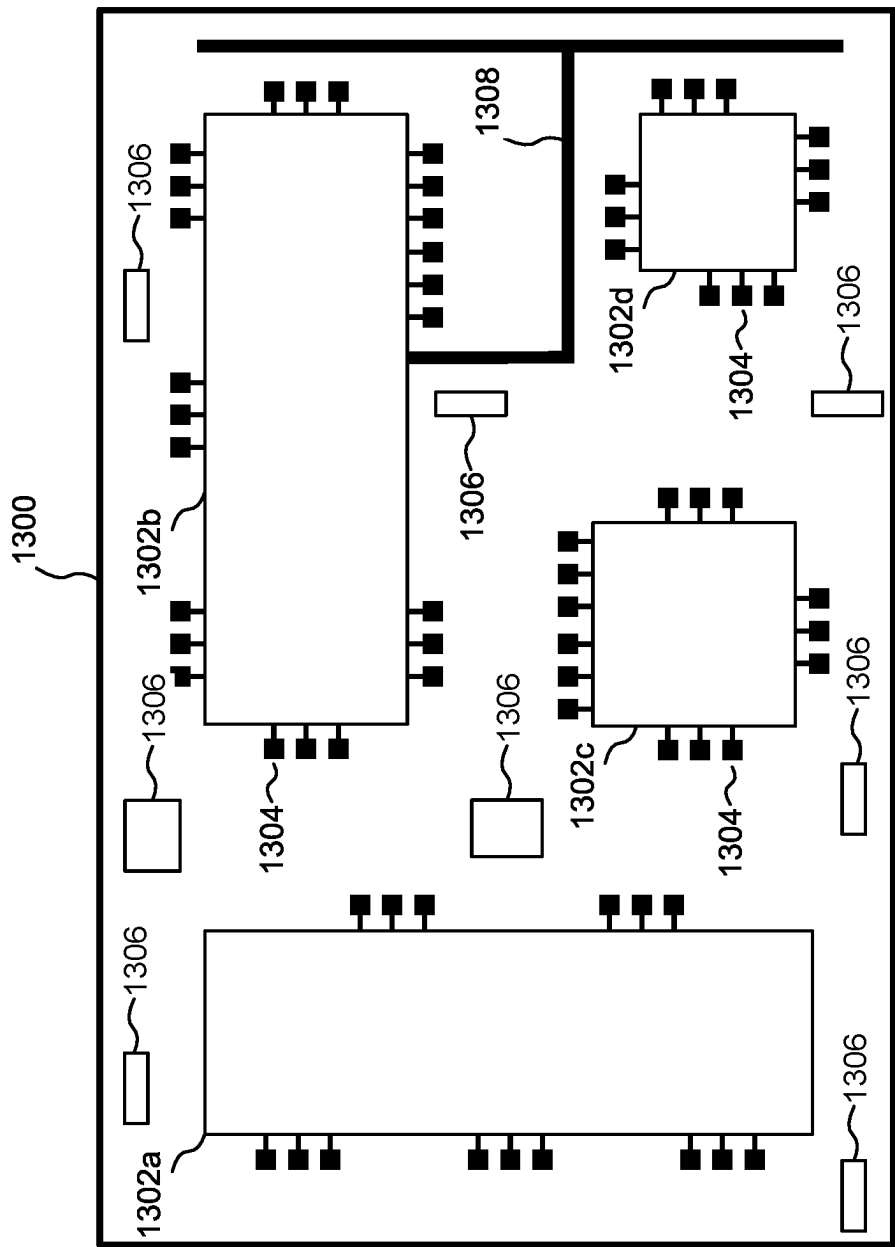


FIG. 13

1400

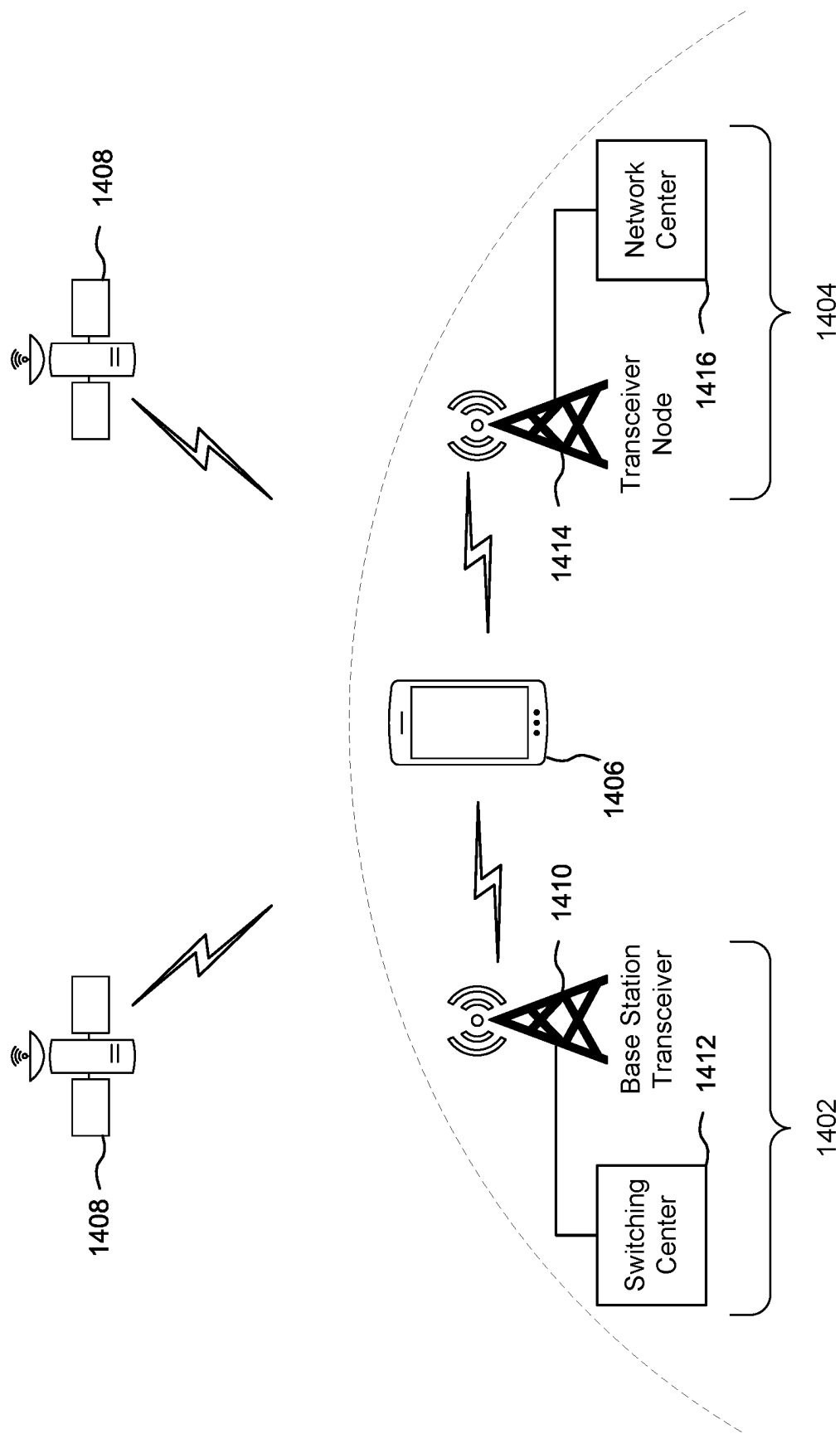


FIG. 14
(Prior Art)

1500

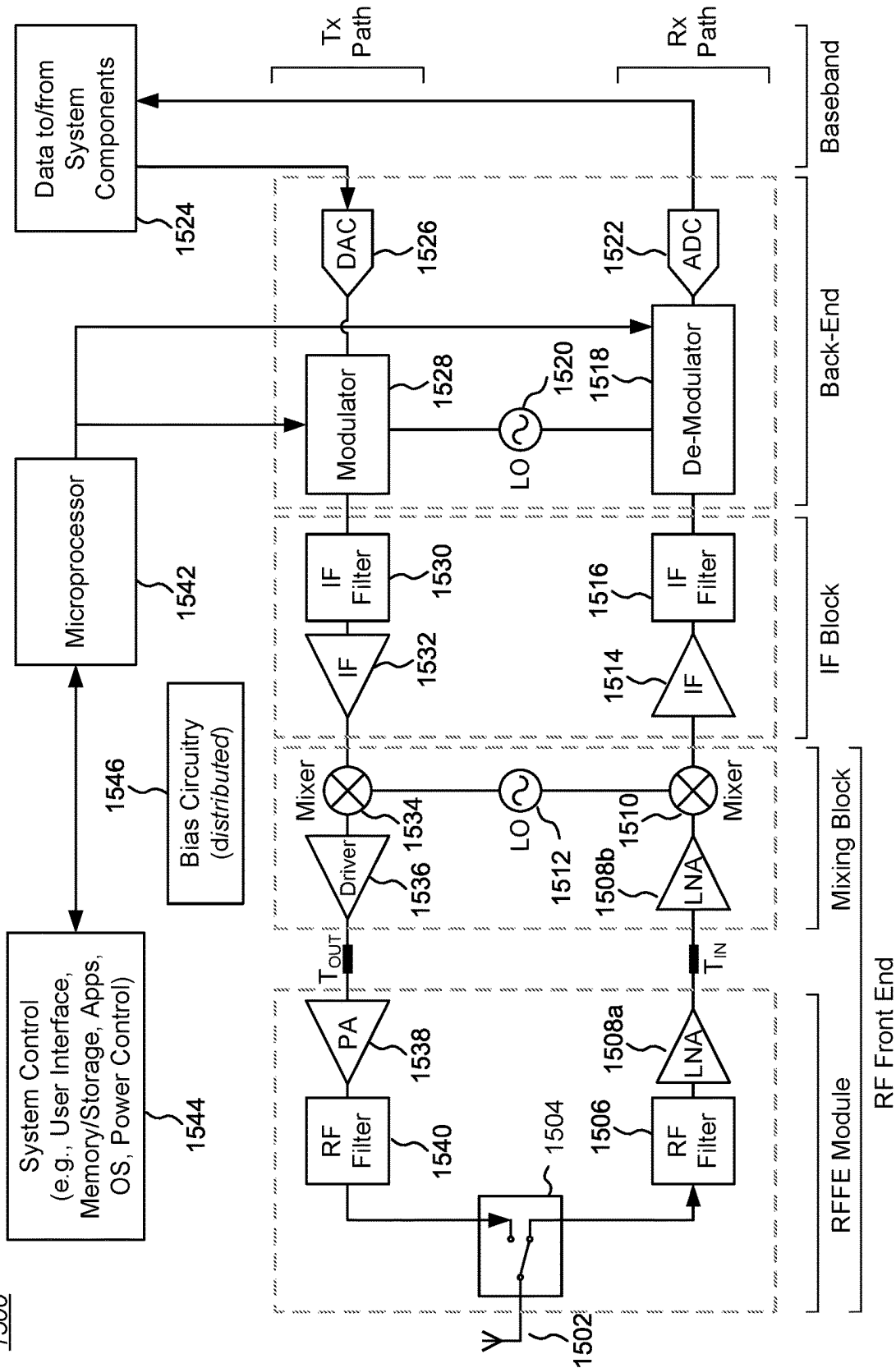
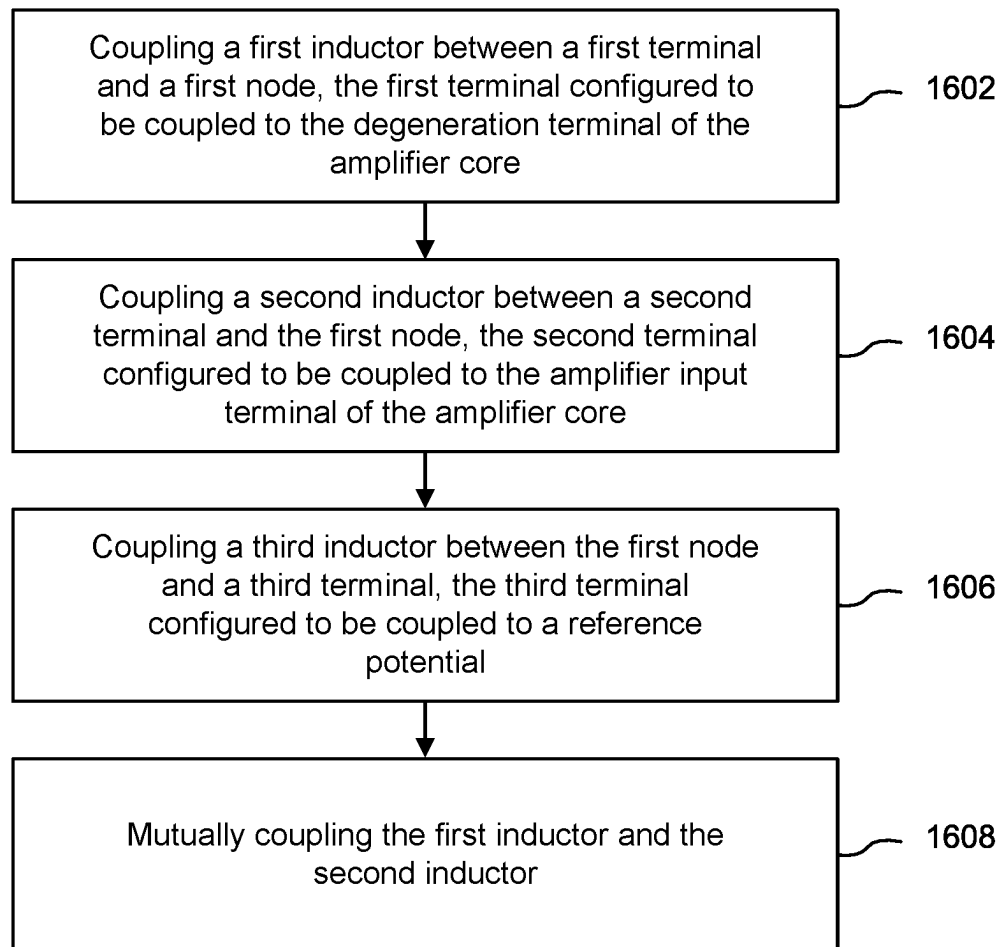
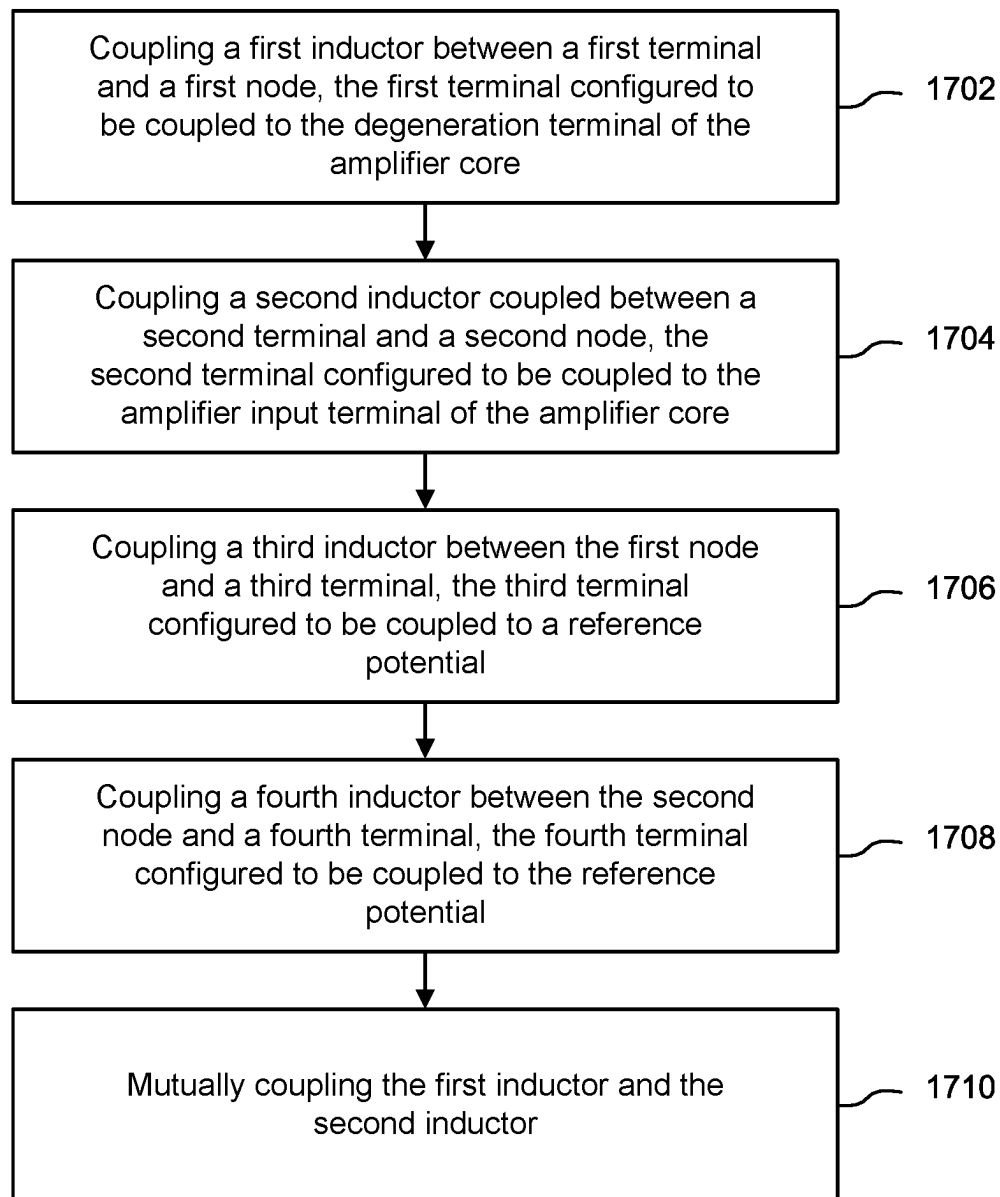


FIG. 15

1600**FIG. 16**

1700**FIG. 17**

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WIDEBAND COUPLED INPUT IMPEDANCE MATCHING LNA ARCHITECTURE

BACKGROUND

(1) Technical Field

The invention relates to electronic circuits, and more particularly to radio frequency amplifier circuits.

(2) Background

Many modern electronic systems include radio frequency (RF) receivers; examples include cellular telephones, personal computers, tablet computers, wireless network components, televisions, cable system “set top” boxes, and radar systems. Many RF receivers are paired with RF transmitters in the form of transceivers, which often are quite complex two-way radios. In some cases, RF transceivers are capable of transmitting and receiving across multiple frequencies in multiple bands.

Amplifiers are a common component in RF transmitters, receivers, and transceivers, and are frequently used for power amplification of transmitted RF signals and for low-noise amplification of received RF signals. For many RF systems, particularly those requiring low power and/or portability (e.g., cellular telephones, WiFi-connected computers, cameras, and other devices), it has become common to use complementary metal-oxide semiconductor (CMOS) fabrication technology to create low cost, low power integrated circuits (ICs). CMOS devices include bulk CMOS, silicon-on-insulator (SOI) CMOS, and silicon-on-sapphire (SOS) CMOS (SOS being a type of SOI fabrication technology).

Receiving an RF signal in many environments requires a high quality low-noise amplifier (LNA) as part of an RF “front end” (RFFE) receiver or transceiver chain of circuits. Important desired characteristics of an LNA are high gain with low noise, a wide bandwidth, good linearity, and good input and output impedance matching. However, in general, all of these factors cannot be optimized simultaneously, and accordingly there are tradeoffs between these characteristics when designing an LNA.

Five important design parameters for LNAs are gain, noise figure (NF), input-referenced third intercept point (IIP3), output reflection coefficient, and input reflection coefficient. NF is a measure of degradation of the signal-to-noise ratio (SNR) caused by components in a signal chain, with lower values indicating better performance. IIP3 is a figure of merit representing amplifier linearity, with higher values indicating better performance. In general, NF has a stricter specification requirement in high-gain modes than in low gain-modes, while IIP3 has a stricter specification requirement in low-gain modes than in high-gain modes. The output reflection coefficient is the S22 scattering parameter (or “S-parameter”) and is an indication of output impedance matching, with lower (more negative, when evaluated logarithmically) numbers indicating better impedance matching. The input reflection coefficient is the S11 S-parameter and is an indication of input impedance matching, with lower (more negative, when evaluated logarithmically) numbers indicating better performance.

Increases in the frequency of RF communications bands and channels, as well as a continuing increase in the number of bands and channels in use, has pushed current LNA architectures to their limits. For example, achieving stringent requirements for gain, percentage bandwidth, linearity,

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and input and output impedance matching with a traditional LNA architecture is difficult, and sometimes impossible, for some of the new 5G mobile network bands, particularly in the 3 to 6 GHz NR bands, the upcoming 7-24 GHz bands, and the millimeter wave range (e.g., 24.25 GHz to 52.6 GHz), owing to insufficient wideband performance.

Wideband LNAs present several design challenges. For example, achieving effective wideband impedance matching at the input of an LNA is a key objective for attaining overall performance, especially NF. Known approaches to input impedance (IM) matching involve using a series inductor or series-inductor/shunt inductor combination between an LNA input terminal and the amplification core (e.g., a cascode transistor pair) but those solutions have limited bandwidth. Multi-stage IM matching comes with other trade-offs, mainly in terms of a lower NF and additional manufacturing cost and integrated circuit (IC) die area.

Accordingly, there is a need for an LNA architecture that overcomes the limitations of conventional LNA architectures.

SUMMARY

The present invention encompasses circuits and methods for a high frequency LNA that include a wideband coupled input impedance matching network. Some embodiments allow multiple modes of operation to allow selection of gain versus linearity and NF characteristics. The inventive circuits and methods may also be applied to other types of amplifiers, such as power amplifiers.

A first embodiment includes a wideband input impedance matching network including: a first inductor coupled between a first terminal and a first node, the first terminal configured to be coupled to a degeneration terminal of an amplifier core; a second inductor coupled between a second terminal and the first node, the second terminal configured to be coupled to an input terminal of the amplifier core; and a third inductor coupled between the first node and a third terminal, the third terminal configured to be coupled to a reference potential; wherein the first inductor and the second inductor are mutually coupled.

A second embodiment includes a wideband input impedance matching network including: a first inductor coupled between a first terminal and a first node, the first terminal configured to be coupled to a degeneration terminal of an amplifier core; a second inductor coupled between a second terminal and a second node, the second terminal configured to be coupled to an input terminal of the amplifier core; a third inductor coupled between the first node and a third terminal, the third terminal configured to be coupled to a reference potential; and a fourth inductor coupled between the second node and a fourth terminal, the fourth terminal configured to be coupled to the reference potential; wherein the first inductor and the second inductor are mutually coupled.

The details of one or more embodiments of the invention are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the invention will be apparent from the description and drawings, and from the claims.

DESCRIPTION OF THE DRAWINGS

FIG. 1 is a simplified schematic diagram of a first embodiment of an LNA circuit having a wideband coupled input impedance matching network.

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FIG. 2 is a simplified schematic diagram of a second embodiment of an LNA circuit having a wideband coupled input impedance matching network.

FIG. 3A is a graph of measured S11 versus frequency performance for an embodiment like LNA of FIG. 1 in comparison to a conventional LNA having a series-inductor/shunt inductor combination between an RF input terminal and an amplification core.

FIG. 3B is a graph of measured gain versus frequency performance for an embodiment like LNA of FIG. 1 in comparison to a conventional LNA having a series-inductor/shunt inductor combination between an RF input terminal and an amplification core.

FIG. 4A is a graph of measured gain versus frequency performance for an embodiment like LNA of FIG. 1 (with common connection of L_S and L_G) in comparison to an embodiment like LNA of FIG. 2 (with separate ground connections for L_S and L_G).

FIG. 4B is a graph of measured S11 versus frequency performance for an embodiment like LNA of FIG. 1 in comparison to an embodiment like LNA of FIG. 2.

FIG. 4C is a graph of measured NF versus frequency performance for an embodiment like LNA of FIG. 1 in comparison to an embodiment like LNA of FIG. 2.

FIG. 5A is a schematic diagram of the LNA from FIG. 1, showing two possible demarcation lines between fabricating WCIIM network circuit components either on-die or off-die (e.g., within a circuit module in which an LNA die is affixed).

FIG. 5B is a schematic diagram of the LNA from FIG. 2, showing two possible demarcation lines between fabricating WCIIM network circuit components either on-die or off-die.

FIG. 6A is a schematic diagram of a portion of the LNA of FIG. 1 showing a second method of achieving a high coupling factor k that is particularly useful when the inductance of L_S is significantly small (e.g., about 0.3 nH) compared to the inductance of L_G (e.g., about 4 nH).

FIG. 6B is a top plan view of an example IC transformer coil in which inductor L_G comprises nearly four turns of the transformer coil, while inductors L_{S1} , L_{S2} , L_{S3} each comprise about one-half of a turn of the transformer coil.

FIG. 7 is a schematic diagram of a portion of the LNA of FIG. 1 showing a switchable configuration for the WCIIM network.

FIG. 8 is a schematic diagram of a portion of the LNA of FIG. 1 showing the WCIIM network coupled to the INT terminal through a series inductance L_{SER} .

FIG. 9A is a graph of simulated gain versus frequency performance for the L_{SER} embodiment in comparison to the non- L_{SER} embodiment.

FIG. 9B is a graph of simulated NF versus frequency performance for the L_{SER} embodiment in comparison to the non- L_{SER} embodiment.

FIG. 9C is a graph of simulated S11 versus frequency performance for the L_{SER} embodiment in comparison to the non- L_{SER} embodiment.

FIG. 9D is a graph of simulated S22 versus frequency performance for the L_{SER} embodiment in comparison to the non- L_{SER} embodiment.

FIG. 9E is a Smith chart of simulated S11 and optimum noise reflection coefficient G_{min} versus frequency performance for the L_{SER} embodiment in comparison to the non- L_{SER} embodiment.

FIG. 9F is a Smith chart of simulated S22 versus frequency performance for the L_{SER} embodiment in comparison to the non- L_{SER} embodiment.

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FIG. 10A is schematic diagram of a first output impedance matching (OIM) network that may be used in conjunction with the LNA circuits of the present disclosure.

FIG. 10B is schematic diagram of a second OIM network that may be used in conjunction with the LNA circuits of the present disclosure.

FIG. 11 is a schematic diagram of one embodiment of an input matching feedback circuit.

FIG. 12 is a simplified schematic diagram of one embodiment of an output matching feedback circuit.

FIG. 13 is a top plan view of a substrate that may be, for example, a printed circuit board or die module substrate (e.g., a thin-film tile).

FIG. 14 illustrates an exemplary prior art wireless communication environment comprising different wireless communication systems, and which may include one or more mobile wireless devices.

FIG. 15 is a block diagram of a transceiver that might be used in a wireless device, such as a cellular telephone, and which may beneficially incorporate an embodiment of the present invention for improved performance.

FIG. 16 is a process flow chart showing a first method for providing wideband input impedance matching for an amplifier core.

FIG. 17 is a process flow chart showing a second method for providing wideband input impedance matching for an amplifier core.

Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

The present invention encompasses circuits and methods for a high frequency LNA that include a wideband coupled input impedance matching network. Some embodiments allow multiple modes of operation to allow selection of gain versus linearity and NF characteristics. The inventive circuits and methods may also be applied to other types of amplifiers, such as power amplifiers.

Bandwidth Terms

For purposes of this disclosure, “narrowband”, “wideband” and “ultra-wideband” may be characterized as a percentage fractional bandwidth equal to (stop frequency f_{STOP} minus start frequency f_{START}) divided by the center frequency f_{CENTER} of a band, or $(f_{STOP}-f_{START})/f_{CENTER}$ (expressed as a percent, where $f_{CENTER}=(f_{STOP}+f_{START})/2$). TABLE 1 below shows typical guidelines (not strict definitions) for characterizing typical percentage bandwidths.

TABLE 1

Nominal Band Name	% bandwidth range
Narrowband	<7.5%
Extended Narrowband	7.5%~15%
Wideband	15%~25%
Ultra-wideband	>25%

TABLE 2 below provides examples of common cellular telephone bands and their characterization as wideband or ultra-wideband using the guidelines in TABLE 1.

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TABLE 2

Typical Application	f_{START} (GHz)	f_{STOP} (GHz)	f_{CENTER} (GHz)	% Bandwidth
Wideband (N77)	3.3	4.2	3.75	24.00%
Wideband (N96)	5.925	7.125	6.525	18.39%
Ultra-Wideband (UWB)	6.2	9	7.6	36.84%

First Embodiment

FIG. 1 is a simplified schematic diagram of a first embodiment of an LNA circuit **100** having a wideband coupled input impedance matching network **102a**.

In the illustrated example, the LNA circuit **100** includes an amplifier core **104** comprising a stack of two series-connected FETs M_{CS} , M_{CG} in a cascode arrangement. An RF input signal applied to an RF input terminal RF_{IN} is coupled through a DC blocking capacitor C_{BLK} to the control gate of the common-source FET M_{CS} , which may be regarded as an input terminal INT of the amplification core **104**.

The source of the common-gate FET M_{CG} is connected to the drain of the common-source FET M_{CS} . The drain of the common-gate FET M_{CG} provides an amplified RF output signal at what may be regarded as an amplified-signal terminal AST of the amplification core **104**.

An output impedance matching (OIM) network **106** has an IN terminal configured to be coupled to the amplified-signal terminal AST of the amplification core **104**, and an OUT terminal configured to be coupled to an RF output terminal R_{OUT} . The amplified output of the amplification core **104** is coupled through the OIM network **106** to the RF output terminal R_{OUT} , which is shown coupled to a typical load represented as a resistor R_L . The value of R_L is typically 50 ohms for many modern RF circuits.

A bias circuit **108** is configured to provide a suitable bias voltage CG_V_{BIAS} to the common-gate FET M_{CG} and a suitable bias voltage CS_V_{BIAS} to the common-source FET M_{CS} , in known fashion. Additional well-known circuit elements that might be included in some applications, such as bypass capacitors, are omitted for clarity.

The wideband coupled input impedance matching (WCIIM) network **102a** includes an RP terminal configured to be coupled to a reference potential (e.g., circuit ground), a DG terminal configured to be coupled to the degeneration terminal DT of the amplification core **104**, and an IN terminal configured to be coupled to the input terminal INT of the amplification core **104** through the capacitor C_{BLK} . In the illustrated embodiment, the WCIIM network **102a** includes a pair of mutually coupled inductors L_S and L_G . Source inductor L_S is coupled between the DG terminal (and thus couplable to the degeneration terminal DT of the amplification core **104**) and a node X, and functions at least in part as a degeneration inductor. Gate inductor L_G is coupled between the IN terminal (and thus couplable to the input terminal INT of the amplification core **104**) and node X. The opposite-side placement of the “dots” adjacent the symbols representing L_S and L_G indicates that the coupling is negative. However, in alternative embodiments, the coupling may be positive—that is, the dots would be on the same sides of the symbols representing L_S and L_G .

The two principal inductors L_G and L_S are laid out such they exhibit a mutual inductance with a coupling factor k_1 and may be implemented, for example, as a coupled segments of a multiport integrated circuit inductor coil. The mutual inductance between L_G and L_S creates a feedback

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current to the input terminal INT of the amplification core **104** which achieves wideband input matching with a minimal impact on NF.

The illustrated WCIIM network **102a** also includes a third inductor L_{R1} symbol coupled between node X (and thus to both L_G and L_S) and the RP terminal (and thus couplable to a reference potential, such as circuit ground). The third inductor L_{R1} represents the inductance inherent to the conductive routing traces from an IC embodiment of the LNA **100** to the ground plane of a module in which the IC is affixed. Since there is control at the design stage of the layout of the conductive routing traces that comprise the third inductor L_{R1} , the inductance value of L_{R1} may be set to further adjust the mutual inductance of L_G and L_S for particular applications.

Second Embodiment

FIG. 2 is a simplified schematic diagram of a second embodiment of an LNA circuit **200** having a wideband coupled input impedance matching network **102b**. The LNA circuit **200** is similar in most respects to the LNA circuit **100** of FIG. 1 but includes a variant WCIIM network **102b**.

The WCIIM network **102b** includes separate RP1 and RP2 terminals configured to be coupled to a reference potential (e.g., circuit ground), a DG terminal configured to be coupled to the degeneration terminal DT of the amplification core **104**, and an IN terminal configured to be coupled to the input terminal INT of the amplification core **104** through the capacitor C_{BLK} . In the illustrated embodiment, the WCIIM network **102b** includes a pair of mutually coupled inductors L_S and L_G . Source inductor L_S is coupled between the DG terminal (and thus couplable to the degeneration terminal DT of the amplification core **104**) and a node X, and functions at least in part as a degeneration inductor. Gate inductor L_G is coupled between the IN terminal (and thus couplable to the input terminal INT of the amplification core **104**) and node Y. The two principal inductors L_G and L_S are laid out such they exhibit a mutual inductance with a coupling factor k_2 and may be implemented, for example, as coupled segments of a multiport integrated circuit inductor coil. The mutual inductance between L_G and L_S creates a feedback current to the input terminal INT of the amplification core **104** which achieves wideband input matching with a minimal impact on NF.

The illustrated WCIIM network **102** also includes a third inductor L_{R1} coupled between node X (and thus to L_S) and the RP1 terminal (and thus couplable to a reference potential, such as circuit ground), and a fourth inductor L_{R2} coupled between node Y (and thus to L_G) and the RP2 terminal (and thus couplable to the reference potential). The third and fourth inductors L_{R1} , L_{R2} represent the inductance inherent to the respective conductive routing traces from an IC embodiment of the LNA **100** to the ground plane of a module in which the IC is affixed. Of note, L_{R1} and L_{R2} are generally not mutually coupled in the illustrated embodiment. Since there is control at the design stage of the layout of the conductive routing traces that comprise the third and fourth inductors L_{R1} , L_{R2} their respective inductance values may be set to further adjust the mutual inductance of L_G and L_S for particular applications.

One challenge in implementing on-die mutually coupled inductors (also known as a transformer) is achieving a transformer having a high coupling factor k ($k=M/\sqrt{L_G/L_S}$, where M is mutual inductance) and a high turn ratio n ($n=\sqrt{L_G/L_S}$). However, by separating the common connection of

L_S and L_G and separately connecting L_S and L_G to the reference potential through L_{R1} and L_{R2} respectively, more design freedom is available for transformer layout to achieve a high k_2 .

Performance Examples

FIG. 3A is a graph 300 of measured S11 versus frequency performance for an embodiment like LNA 100 of FIG. 1 in comparison to a conventional LNA having a series-inductor/shunt inductor combination between an RF input terminal and an amplification core 104. Graph line 302 corresponds to the LNA 100, while graph line 304 corresponds to the conventional LNA. Both embodiments were designed for optimal performance between about 6 GHz and about 9 GHz (that range is indicated by dashed lines 306 and 308); as examination of graph 300 shows, the LNA 100 architecture exhibits better average S11 performance in that frequency interval.

FIG. 3B is a graph 310 of measured gain versus frequency performance for an embodiment like LNA 100 of FIG. 1 in comparison to a conventional LNA having a series-inductor/shunt inductor combination between an RF input terminal and an amplification core 104. Graph line 312 corresponds to the LNA 100, while graph line 314 corresponds to the conventional LNA. In the 6-9 GHz frequency interval, the LNA 100 architecture exhibits flatter gain performance with better minimum gain in the example frequency interval.

As between the LNA 100 and the conventional LNA, the NF performance (not shown) is comparable, with the LNA 100 generally exhibiting somewhat flatter performance and better average performance in the example frequency interval.

FIG. 4A is a graph 400 of measured gain versus frequency performance for an embodiment like LNA 100 of FIG. 1 (with common connection of L_S and L_G) in comparison to an embodiment like LNA 200 of FIG. 2 (with separate ground connections for L_S and L_G). Graph line 402 corresponds to the LNA 100 embodiment, while graph line 404 corresponds to the LNA 200 embodiment, which exhibits higher gain over the example frequency interval (corresponding to the 5G N77 band).

FIG. 4B is a graph 410 of measured S11 versus frequency performance for an embodiment like LNA 100 of FIG. 1 in comparison to an embodiment like LNA 200 of FIG. 2. Graph line 412 corresponds to the LNA 100 embodiment, while graph line 414 corresponds to the LNA 200 embodiment, which exhibits somewhat higher S11 over the example frequency interval.

FIG. 4C is a graph 420 of measured NF versus frequency performance for an embodiment like LNA 100 of FIG. 1 in comparison to an embodiment like LNA 200 of FIG. 2. Lower NF values indicate better performance. Graph line 422 corresponds to the LNA 100 embodiment, while graph line 424 corresponds to the LNA 200 embodiment, which exhibits somewhat higher average NF over the example frequency interval.

In this example, the LNA 100 needs more coupling on the IC to offset the opposite coupling on the module; the separate ground connections of the LNA 200 overcomes that issue and achieves higher gain, but with some trade-offs with respect to S11 and NF performance. It should be appreciated that either architecture may be used for any particular application, with performance trade-offs among gain, S11, and NF as may be needed to meet an applicable specification.

Compared to input impedance matching circuits having only a series-inductor or a series-inductor/shunt inductor combination between an LNA input terminal and the amplification core, the new impedance matching input architectures described in this disclosure have the following benefits: IC area savings, manufacturing cost savings, wider impedance matching bandwidth, and flatter gain response.

IC to Module Connection and Mutual Coupling Options

Integrated circuit embodiments of the LNA circuits 100, 200 may take advantage of a number of different possible circuit layouts for trading-off inductor coupling and IC area. For example, FIG. 5A is a schematic diagram of the LNA 100 from FIG. 1, showing two possible demarcation lines between fabricating WCIIM network 102a circuit components either on-die or off-die (e.g., within a circuit module in which an LNA die is affixed) with respect to the amplifier core 104. In a first configuration 502, the mutually-coupled inductors L_G , L_S and the common inductor L_{R1} are located off-die with respect to the amplifier core 104, thus saving IC area. In a second configuration 504, the mutually-coupled inductors L_G , L_S are fabricated on-die (i.e., co-fabricated with the amplifier core 104), while the common inductor L_{R1} (which essentially is always present due to routing requirements) is located off-die with respect to the amplifier core 104. In the second illustrated configuration, coupling is negative on the IC and positive off-die (e.g., in the module). Having the coupling on-die generally gives more control over the design characteristics.

As another example, FIG. 5B is a schematic diagram of the LNA 200 from FIG. 2, showing two possible demarcation lines between fabricating WCIIM network 102b circuit components either on-die or off-die. In a first configuration 512, the mutually-coupled inductors L_G , L_S and the separate inductors L_{R1} , L_{R2} are fabricated off-die, thus saving IC area. In a second configuration 514, the mutually-coupled inductors L_G , L_S are fabricated on-die, while the separate inductors L_{R1} , L_{R2} are fabricated off-die. In the second illustrated configuration, coupling is negative on the IC and negligible off-die (e.g., in the module). Having the coupling on-die generally gives more control over the design characteristics.

Alternative Embodiments

The present invention encompasses a number of alternative embodiments. For example, FIG. 6A is a schematic diagram of a portion 600 of the LNA 100 of FIG. 1 showing a second method of achieving a high coupling factor k that is particularly useful when the inductance of L_S is significantly small (e.g., about 0.3 nH) compared to the inductance of L_G (e.g., about 4 nH). As shown in FIG. 6A, the single inductor L_S of FIG. 1 is replaced by multiple parallel inductors L_{S1} , L_{S2} , . . . L_{Sn} , where $n \geq 2$. The multiple parallel inductors may be implemented using mutually-coupled parallel single-turn inductors instead of using a conventional multi-turn single inductor. For example, FIG. 6B is a top plan view of an example IC transformer coil 610 in which inductor L_G comprises nearly four turns of the transformer coil 610, while inductors L_{S1} , L_{S2} , L_{S3} each comprise about one-half of a turn of the transformer coil 610. Note that in the region encompassed by the dashed oval 612, the conductive traces of inductors L_{S1} , L_{S2} , L_{S3} overlay the conductive traces of inductor L_G . The resulting layout of the coupled inductors within the WCIIM network 102a is beneficially compact for IC fabrication. The configuration of

multiple parallel inductors $L_{S1}, L_{S2}, \dots, L_{Sn}$ may also be used in conjunction with the LNA 200 of FIG. 2. The concepts shown in FIGS. 6A and 6B may also be applied to the LNA 200 of FIG. 2, along with comparable LNAs.

FIG. 7 is a schematic diagram of a portion 700 of the LNA 100 of FIG. 1 showing a switchable configuration for the WCIIM network 102a. In the illustrated embodiment, the source inductor L_S of FIG. 1 is “split” to include two transformer coils, L_{Sa} and L_{Sb} that can be switchable coupled to inductor L_G . Also included are a set of switches that enable different configurations of the WCIIM network 102a—more specifically, a first switch Sw1 coupled between the degeneration terminal DT and circuit ground (as one example of a reference voltage), a second switch Sw2 coupled between node X and circuit ground, a third switch Sw3 coupled between inductor L_{Sb} and circuit ground, and a fourth switch Sw4 coupled between inductor L_G and circuit ground. Omitted to avoid clutter is the inductance inherent to the conductive routing traces from an IC embodiment of the LNA 100 to the ground plane of a module in which the IC is affixed. The concepts shown in FIG. 7 may also be applied to the LNA 200 of FIG. 2, along with comparable LNAs.

The illustrated architecture may be configured for different modes that trade off gain against linearity. For example, TABLE 3 below provides just a few examples of switch positions and resulting modes achievable with the switchable configuration for the WCIIM network 102a. In the table, “O” means a switch is set to an OPEN (non-conducting) state, “X” means a switch is set to a CLOSED (conducting) state, and “-” means a switch may be set to OPEN or CLOSED as desired for a particular application. As should be appreciated, other combinations of switch configurations are possible.

TABLE 3

Mode	Sw1	Sw2	Sw3	Sw4	Sw5	Observations
High gain	O	X	O	X	O	Inductors L_G and L_{Sa} are coupled with a coupling factor k1.
Low gain	O	O	X	X	O	Inductors L_G and $L_{Sa} + L_{Sb}$ (i.e., the series sum of the inductances) are coupled with a coupling factor k2. Depending on the value and layout of L_{Sb} , the turn ratio and coupling factor can be adjusted for different gain and linearity tradeoffs. This is especially useful in a very low current and very low gain mode, where linearity is more important.
Alt. Mode 1	—	—	—	O	X	Inductor L_G is essentially deactivated (electrically disconnected), with no feedback connection to the INT terminal.
Alt. Mode 2	X	O	O	X	O	Inductors $L_{Sa} + L_{Sb}$ are essentially deactivated (bypassed)
Alt. Mode 3	X	O	O	O	X	Inductor L_G is essentially deactivated (electrically disconnected), with no feedback connection to the INT terminal, and inductors $L_{Sa} + L_{Sb}$ are essentially deactivated (bypassed)

Depending on different application scenarios, switch Sw3 may be omitted (thus connecting L_{Sb} to circuit ground) for different tradeoffs. For example, in a first topology, if low-loss switches are available and the Q of L_{Sa} is important, then switch Sw3 may be included in series with L_{Sb} as shown in FIG. 7. This helps improve the Q of L_{Sa} when Sw2 is CLOSED and Sw3 is OPEN, since Sw3 prevents signal from flowing through L_{Sb} (noting that actual FET implementations of Sw2 will not present as an ideal short with zero resistance, so closing Sw2 would not completely

bypass L_{Sb}). In a second topology, if low-loss switches are not available, Sw3 may be omitted. This helps improve the Q of the serial combination of $L_{Sa} + L_{Sb}$ when Sw2 is OPEN. The configuration of a split, switchable L_S inductor may also be used in conjunction with the LNA 200 of FIG. 2.

FIG. 8 is a schematic diagram of a portion 800 of the LNA 100 of FIG. 1 showing the WCIIM network 102a coupled to the INT terminal through a series inductance L_{SER} and a DC blocking capacitor C_{BLK} . More specifically, the inductor L_{SER} is coupled to the input terminal INT of the amplification core 104 through a DC blocking capacitor C_{BLK} (the location of C_{BLK} and L_{SER} can be swapped), and to the inductor L_G of the WCIIM network 102a such that feedback from the WCIIM network 102a flows through L_{SER} . This architecture helps input impedance matching with an additional degree of freedom and improves both gain and NF. A series inductance L_{SER} may also be used in conjunction with the LNA 200 of FIG. 2, along with comparable LNAs.

For example, FIGS. 9A-9E are graphs of various LNA parameter values as a function of frequency for an embodiment like LNA 100 of FIG. 1 (with common connection of L_S and L_G) where the LNA includes a WCIIM network 102a coupled to the INT terminal through a series inductance L_{SER} and a DC blocking capacitor C_{BLK} (the “ L_{SER} embodiment”) and for a similar LNA embodiment that lacks L_{SER} ((the “non- L_{SER} embodiment”). The LNA embodiments in this example are designed for the 5-7 GHz 5G New Radio Unlicensed (NR-U) cellular radio spectrum. The illustrated graphs show that gain, S11, S22, and NF can be improved concurrently with the inclusion of the L_{SER} inductance.

FIG. 9A is a graph 900 of simulated gain versus frequency performance for the L_{SER} embodiment (graph line 902) in comparison to the non- L_{SER} embodiment (graph line 904). An increase in the series inductance at the input improves

the Q of the input matching network, and thus improves overall LNA gain. Accordingly, the L_{SER} embodiment exhibits higher gain over the example frequency interval.

FIG. 9B is a graph 910 of simulated NF versus frequency performance for the L_{SER} embodiment (graph line 912) in comparison to the non- L_{SER} embodiment (graph line 914). Lower NF values indicate better performance. The L_{SER} embodiment exhibits lower NF over the example frequency interval.

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FIG. 9C is a graph **920** of simulated S11 versus frequency performance for the L_{SER} embodiment (graph line **922**) in comparison to the non- L_{SER} embodiment (graph line **924**). Lower S11 values indicate better performance. The L_{SER} embodiment exhibits lower S11 over the example frequency interval.

FIG. 9D is a graph **930** of simulated S22 versus frequency performance for the L_{SER} embodiment (graph line **932**) in comparison to the non- L_{SER} embodiment (graph line **934**). Lower S22 values indicate better performance. The L_{SER} embodiment exhibits lower S22 over the example frequency interval.

FIG. 9E is a Smith chart **940** of simulated S11 and optimum noise reflection coefficient G_{min} versus frequency performance for the L_{SER} embodiment (graph lines **942**_{S11} and **942**_{Gmin}) in comparison to the non- L_{SER} embodiment (graph lines **944**_{S11} and **944**_{Gmin}). As the Smith chart **940** shows, the inclusion of the L_{SER} inductance causes the S11 (graph line **942**_{S11}) for the L_{SER} embodiment to be closer to the center **946** of the chart (indicating better S11 performance than the non- L_{SER} embodiment). Concurrently, the inclusion of the L_{SER} inductance causes the G_{min} (graph line **942**_{Gmin}) for the L_{SER} embodiment to be closer to the center **946** of the chart (indicating better noise matching than the non- L_{SER} embodiment).

FIG. 9F is a Smith chart **950** of simulated S22 versus frequency performance for the L_{SER} embodiment (graph line **952**) in comparison to the non- L_{SER} embodiment (graph line **954**). As the Smith chart **950** shows, the inclusion of the L_{SER} inductance places the S22 (graph line **952**) for the L_{SER} embodiment closer to the center **956** of the chart (indicating better S22 performance than the non- L_{SER} embodiment).

Optional Circuit Elements

A. Output Impedance Matching Circuits. The OIM network **106** of the above-described LNA circuits may be implemented in a number of ways that contribute to wideband operation.

For example, FIG. **10A** is schematic diagram of a first output impedance matching (OIM) network **106a** that may be used in conjunction with the LNA circuits **100**, **200** of the present disclosure. The OIM network **106a** includes a first inductor **L1** coupled in series with a second inductor **L2** at a node **Z**, with the pair of inductors **L1**, **L2** coupled to the IN terminal of the OIM network **106a** and configured to be coupled to a power supply V_{DD} . A DC blocking capacitor C_{BLK1} is coupled between node **Z** and the OUT terminal of the OIM network **106a**, which in turn is coupled to the R_{OUT} terminal. Any of the inductors **L1**, **L2** and capacitor C_{BLK1} may be adjustable components, as shown, or have fixed values.

As another example, FIG. **10B** is schematic diagram of a second OIM network **106b** that may be used in conjunction with the LNA circuits **100**, **200** of the present disclosure. The OIM network **106b** includes an inductor **L2** coupled to the IN terminal of the OIM network **106b** and configured to be coupled to a power supply V_{DD} . A DC blocking capacitor C_{BLK2} is coupled between the amplified-signal terminal **AST** of the amplification core **104** and the control gate of a second-stage FET M_{SF} , which provides a second stage of amplification as a source-follower circuit. In the illustrated example, the second-stage FET M_{SF} has a conduction channel (drain to source) coupled between V_{DD} and an inductor **L3**, which in turn is coupled to a reference potential, such as circuit ground. The source of the second-stage FET M_{SF} is coupled through a DC blocking capacitor C_{BLK3} to the OUT

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terminal of the OIM network **106b**, which in turn is coupled to the RF_{OUT} terminal. Some embodiments may include a capacitor C_{F0} coupled in parallel with the inductor **L3**. Any of the inductors **L2**, **L3** and capacitors C_{BLK2} , C_{BLK3} , C_{F0} may be adjustable components, as shown, or have fixed values.

B. Stack of FETS. In some embodiments, in order to overcome a relatively low breakdown voltage per CMOS FET, multiple common-gate FETS may be series-coupled in a FET stack **120** between the drain of the bottom-most common-gate FET M_{CG} and the amplified-signal terminal **AST**, as shown in FIGS. **1** and **2**—that is, the amplification core **104** may have multiple series-coupled common-gate FETs in a cascode configuration. The amplified-signal terminal **AST** would then be at the drain of the upper-most common-gate FET in the amplification core **104**.

C. Input Matching Feedback Circuit. Some embodiments may include an input matching (IM) feedback circuit **122**. The IM feedback circuit **122** is shown in FIGS. **1** and **2** as coupled between the input terminal **INT** and the amplified-signal terminal **AST**. More generally, the IM feedback circuit **122** may be coupled to a feedback node in the output signal path of the amplification core **104**, which may be the drain of any of the FETS in the amplification core **104** (including within a FET stack **120**). The choice of feedback node for connection to the IM feedback circuit **122** may be made, for example, based on different desired feedback strength and different desired input impedance.

FIG. **11** is a schematic diagram of one embodiment of an input matching feedback circuit **122**. The IM feedback circuit **122** in the illustrated example comprises a DC blocking capacitor C_{F1} series-coupled with a switch S_{F1} , which in turn is series-coupled to a resistor R_{F1} . In some embodiments, C_{F1} and/or R_{F1} may be adjustable, as shown, or have fixed values. In alternative embodiments, the IM feedback circuit **122** may include one or more parallel switches $S_{F11} \dots S_{F1n}$ (generically, S_{F1x}), which in turn are series-coupled to respective parallel resistor(s) $R_{F11} \dots R_{F1n}$ (generically, R_{F1x}). With n parallel switch/resistor branches, the tunability of the resistors R_{F1x} is equal to 2^n (the total number of switching combinations). Note that the values of the resistors R_{F1x} need not be identical. The IM feedback circuit **122** may be disabled by opening all feedback switches S_{F1x} .

In alternative embodiments, capacitor C_{F1} , the set of one or more switches S_{F1x} and the set of one or more resistors R_{F1x} may be connected in any series order. In still other embodiments, the switches S_{F1x} may be omitted, thereby permanently coupling the set of one or more resistors R_{F1x} between the input terminal **INT** and a feedback node in the output signal path of the amplification core **104**.

In various gain modes, the in-circuit presence of one or more resistors R_{F1x} allows the Q-factor of the input impedance matching to be reduced, which increases the bandwidth of the LNA circuit at the expense of gain and NF.

An advantage of using a variable or multi-state IM feedback circuit **122** is that multiple resistance values enable multiple gain modes. For instance, LNAs in mobile RF receiver devices may need multiple gain modes depending upon the range of input signal strength at the receiver. In addition, enabling multiple gain modes by using variable or multi-state feedback resistors R_{F1x} may eliminate the need for an output attenuator (common in conventional receiver LNAs).

Further information regarding the IM feedback circuit **122** may be found in U.S. patent application Ser. No. 17/337, 227, filed Jun. 2, 2021, entitled "Wideband Multi Gain LNA

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Architecture”, assigned to the assignee of the present invention, the contents of which are hereby incorporated by reference.

D. Output Matching Feedback Circuit. Some embodiments may include an output matching (OM) deQing circuit **124**. The OM deQing circuit **124** is shown in FIGS. **1** and **2** as coupled between the amplified-signal terminal AST and the control gate of the common-gate FET M_{CG} , and functions to lower the transformation Q of the OIM network **106**. More generally, the OM deQing circuit **124** may be coupled to a feedback node in the output signal path of the amplification core **104**, which may be the drain of any of the FETs in the amplification core **104** (including within a FET stack **120**). Lowering the transformation Q extends the output impedance matching bandwidth of the LNA circuit. This architecture shows a better gain and bandwidth tradeoff in a wide variety of applications compared to other known circuits.

FIG. **12** is a simplified schematic diagram of one embodiment of an output matching deQing circuit **124**. The OM deQing circuit **124** in the illustrated example comprises a DC blocking capacitor C_{F2} series-coupled with a switch S_{F2} , which in turn is series-coupled to a resistor R_{F2} . In some embodiments, C_{F2} and/or R_{F2} may be adjustable, as shown, or have fixed values. In alternative embodiments, the OM deQing circuit **124** may include one or more parallel switches $S_{F21} \dots S_{F2n}$ (generically, S_{F2x}), which in turn are series-coupled to respective parallel resistor(s) $R_{F21} \dots R_{F2n}$ (generically, R_{F2x}). With n parallel switch/resistor branches, the tunability of the resistors R_{F2x} is equal to 2^n (the total number of switching combinations). Note that the values of the resistors R_{F2x} need not be identical. The OM deQing circuit **124** may be disabled by opening all feedback switches S_{F2x} .

In alternative embodiments, capacitor C_{F2} , the set of one or more switches S_{F2x} , and the set of one or more resistors R_{F2x} be connected in any series order. In still other embodiments, the switches S_{F2x} may be omitted, thereby permanently coupling the set of one or more resistors R_{F2x} between the amplified-signal terminal AST and the control gate of the common-gate FET M_{CG} .

In various gain modes, the in-circuit presence of one or more resistors R_{F2x} allows the Q-factor of the output impedance matching to be reduced, which increases the bandwidth of the LNA circuit at the expense of gain and NF. For example, switching a single resistor R_{F2} in-circuit or out-of-circuit provides two operation modes, a first mode having a wider output impedance matching bandwidth, and a second mode having a narrower output impedance matching bandwidth but with higher gain than the first mode (assuming that no other LNA circuit elements are concurrently adjusted). As noted above, making R_{F2} variable or enabling more than one resistor value enables additional modes of operation.

More specifically, when switch S_{F2} is CLOSED, R_{F2} couples the amplified-signal terminal AST to the control gate of the common-gate FET M_{CG} , effectively placing R_{F2} in parallel with the equivalent resistance R_D of the amplifier core **104** circuit. The in-circuit presence of R_{F2} lowers the impedance, Z_{DRAIN} , of the amplifier core **104** as seen at the AST terminal, and reduces the transformation Q of the OIM network **106**, where Q is approximately equal to the real part of Z_{DRAIN} divided by the load resistance R_L , or $\text{Re}(Z_{DRAIN})/R_L$. Accordingly, the transformation Q is lowered, thereby extending the output impedance matching bandwidth of the LNA circuit. This architecture shows a better gain and bandwidth tradeoff in a wide variety of applications com-

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pared to other known circuits. Conversely, when switch S_{F2} is OPEN, R_{F2} is out-of-circuit with respect to R_D in the equivalent circuit, and the transformation Q of the OIM network **106** is not reduced.

An advantage of using a variable or multi-state OM deQing circuit **124** is that multiple resistance values enable multiple gain modes. For instance, LNAs in mobile RF receiver devices may need multiple gain modes depending upon the range of input signal strength at the receiver. In addition, enabling multiple gain modes by using variable or multi-state feedback resistors R_{F2x} may eliminate the need for an output attenuator (common in conventional receiver LNAs).

Further information regarding the OM deQing circuit **124** may be found in U.S. patent application Ser. No. 17/855,386, filed concurrently herewith, entitled “Extended Impedance Matching Wideband LNA Architectures”, assigned to the assignee of the present invention, the contents of which are hereby incorporated by reference.

Circuit Embodiments

Circuits and devices in accordance with the present invention may be used alone or in combination with other components, circuits, and devices. For example, the LNA circuits of FIGS. **1**, **2**, **6A**, **7**, and **8** may be used with either of the output impedance matching networks **106a**, **106b** of FIGS. **10A** and **10B**, respectively, and with or without the IM feedback circuit **122** of FIG. **11** and/or the output matching feedback circuit **124** of FIG. **12**.

Embodiments of the present invention may be fabricated as integrated circuits (ICs), which may be encased in IC packages and/or in modules for ease of handling, manufacture, and/or improved performance. In particular, IC embodiments of this invention are often used in modules in which one or more of such ICs are combined with other circuit components or blocks (e.g., filters, amplifiers, passive components, and possibly additional ICs) into one package. The ICs and/or modules are then typically combined with other components, often on a printed circuit board, to form part of an end product such as a cellular telephone, laptop computer, or electronic tablet, or to form a higher-level module which may be used in a wide variety of products, such as vehicles, test equipment, medical devices, etc. Through various configurations of modules and assemblies, such ICs typically enable a mode of communication, often wireless communication.

As one example of further integration of embodiments of the present invention with other components, FIG. **13** is a top plan view of a substrate **1300** that may be, for example, a printed circuit board or die module substrate (e.g., a thin-film tile). In the illustrated example, the substrate **1300** includes multiple ICs **1302a-1302d** having terminal pads **1304** which would be interconnected by conductive vias and/or traces on and/or within the substrate **1300** or on the opposite (back) surface of the substrate **1300** (to avoid clutter, the surface conductive traces are not shown and not all terminal pads are labelled). The ICs **1302a-1302d** may embody, for example, signal switches, active filters, amplifiers (including one or more LNAs), and other circuitry. For example, IC **1302b** may incorporate one or more instances of an LNA like the LNA circuits **100**, **200** shown in FIGS. **1** and **2**.

The substrate **1300** may also include one or more passive devices **1306** embedded in, formed on, and/or affixed to the substrate **1300**. While shown as generic rectangles, the passive devices **1306** may be, for example, filters, capaci-

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tors, inductors, transmission lines, resistors, planar antennae elements, transducers (including, for example, MEMS-based transducers, such as accelerometers, gyroscopes, microphones, pressure sensors, etc.), batteries, etc., interconnected by conductive traces on or in the substrate **1300** to other passive devices **1306** and/or the individual ICs **1302a-1302d**.

The front or back surface of the substrate **1300** may be used as a location for the formation of other structures. For example, one or more antennae may be formed on or affixed to the front or back surface of the substrate **1300**; one example of a front-surface antenna **1308** is shown, coupled to an IC die **1302b**, which may include RF front-end circuitry. Thus, by including one or more antennae on the substrate **1300**, a complete radio may be created.

System Aspects

Embodiments of the present invention are useful in a wide variety of larger radio frequency (RF) circuits and systems for performing a range of functions, including (but not limited to) impedance matching circuits, RF power amplifiers, RF low-noise amplifiers (LNAs), phase shifters, attenuators, antenna beam-steering systems, charge pump devices, RF switches, etc. Such functions are useful in a variety of applications, such as radar systems (including phased array and automotive radar systems), radio systems (including cellular radio systems), and test equipment.

Radio system usage includes wireless RF systems (including base stations, relay stations, and hand-held transceivers) that use various technologies and protocols, including various types of orthogonal frequency-division multiplexing ("OFDM"), quadrature amplitude modulation ("QAM"), Code-Division Multiple Access ("CDMA"), Time-Division Multiple Access ("TDMA"), Wide Band Code Division Multiple Access ("W-CDMA"), Global System for Mobile Communications ("GSM"), Long Term Evolution ("LTE"), 5G, 6G, and WiFi (e.g., 802.11a, b, g, ac, ax, be) protocols, as well as other radio communication standards and protocols.

As an example of wireless RF system usage, FIG. **14** illustrates an exemplary prior art wireless communication environment **1400** comprising different wireless communication systems **1402** and **1404**, and which may include one or more mobile wireless devices **1406**. A wireless device **1406** may be a cellular phone, a wireless-enabled computer or tablet, or some other wireless communication unit or device. A wireless device **1406** may also be referred to as a mobile station, user equipment, an access terminal, or some other terminology known in the telecommunications industry.

A wireless device **1406** may be capable of communicating with multiple wireless communication systems **1402**, **1404** using one or more of telecommunication protocols such as the protocols noted above. A wireless device **1406** also may be capable of communicating with one or more satellites **1408**, such as navigation satellites (e.g., GPS) and/or telecommunication satellites. The wireless device **1406** may be equipped with multiple antennas, externally and/or internally, for operation on different frequencies and/or to provide diversity against deleterious path effects such as fading and multi-path interference.

The wireless communication system **1402** may be, for example, a CDMA-based system that includes one or more base station transceivers (BSTs) **1410** and at least one switching center (SC) **1412**. Each BST **1410** provides over-the-air RF communication for wireless devices **1406** within

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its coverage area. The SC **1412** couples to one or more BSTs **1410** in the wireless system **1402** and provides coordination and control for those BSTs **1410**.

The wireless communication system **1404** may be, for example, a TDMA-based system that includes one or more transceiver nodes **1414** and a network center (NC) **1416**. Each transceiver node **1414** provides over-the-air RF communication for wireless devices **1406** within its coverage area. The NC **1416** couples to one or more transceiver nodes **1414** in the wireless system **1404** and provides coordination and control for those transceiver nodes **1414**.

In general, each BST **1410** and transceiver node **1414** is a fixed station that provides communication coverage for wireless devices **1406**, and may also be referred to as base stations or some other terminology known in the telecommunications industry. The SC **1412** and the NC **1416** are network entities that provide coordination and control for the base stations and may also be referred to by other terminologies known in the telecommunications industry.

An important aspect of any wireless system, including the systems shown in FIG. **14**, is in the details of how the component elements of the system perform. FIG. **15** is a block diagram of a transceiver **1500** that might be used in a wireless device, such as a cellular telephone, and which may beneficially incorporate an embodiment of the present invention for improved performance. As illustrated, the transceiver **1500** includes a mix of RF analog circuitry for directly conveying and/or transforming signals on an RF signal path, non-RF analog circuitry for operational needs outside of the RF signal path (e.g., for bias voltages and switching signals), and digital circuitry for control and user interface requirements. In this example, a receiver path Rx includes RF Front End, Intermediate Frequency (IF) Block, Back-End, and Baseband sections (noting that in some implementations, the differentiation between sections may be different). The various illustrated sections and circuit elements may be embodied in one die or multiple IC dies. For example, the RF Front End in the illustrated example may include an RFFE module and a Mixing Block, which may be embodied in (or as part of) different IC dies or modules. The different dies and/or modules may be coupled by transmission lines T_{IN} and T_{OUT} (e.g., microstrips, coplanar waveguides, or an equivalent structure or circuit), either or both of which may have, for example, a 50Ω impedance.

The receiver path Rx receives over-the-air RF signals through at least one antenna **1502** and a switching unit **1504**, which may be implemented with active switching devices (e.g., field effect transistors or FETs) and/or with passive devices that implement frequency-domain multiplexing, such as a diplexer or duplexer. An RF filter **1506** passes desired received RF signals to at least one low noise amplifier (LNA) **1508a**, the output of which is coupled from the RFFE Module to at least one LNA **1508b** in the Mixing Block (through transmission line T_{IN} in this example). The LNA(s) **1508b** may provide buffering, input matching, and reverse isolation. The LNAs **1508a**, **1508b** may be instances of any of the LNAs disclosed in FIGS. **1**, **2**, **6A**, **7**, and **8**.

The output of the LNA(s) **1508b** is combined in a corresponding mixer **1510** with the output of a first local oscillator **1512** to produce an IF signal. The IF signal may be amplified by an IF amplifier **1514** and subjected to an IF filter **1516** before being applied to a demodulator **1518**, which may be coupled to a second local oscillator **1520**. The demodulated output of the demodulator **1518** is transformed to a digital signal by an analog-to-digital converter **1522** and provided to one or more system components **1524** (e.g., a

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video graphics circuit, a sound circuit, memory devices, etc.). The converted digital signal may represent, for example, video or still images, sounds, or symbols, such as text or other characters.

In the illustrated example, a transmitter path Tx includes Baseband, Back-End, IF Block, and RF Front End sections (again, in some implementations, the differentiation between sections may be different). Digital data from one or more system components **1524** is transformed to an analog signal by a digital-to-analog converter **1526**, the output of which is applied to a modulator **1528**, which also may be coupled to the second local oscillator **1520**. The modulated output of the modulator **1528** may be subjected to an IF filter **1530** before being amplified by an IF amplifier **1532**. The output of the IF amplifier **1532** is then combined in a mixer **1534** with the output of the first local oscillator **1512** to produce an RF signal. The RF signal may be amplified by a driver **1536**, the output of which is coupled to a power amplifier (PA) **1538** (through transmission line T_{OUT} in this example). The amplified RF signal may be coupled to an RF filter **1540**, the output of which is coupled to at least one antenna **1502** through the switching unit **1504**.

The operation of the transceiver **1500** is controlled by a microprocessor **1542** in known fashion, which interacts with system control components **1544** (e.g., user interfaces, memory/storage devices, application programs, operating system software, power control, etc.). In addition, the transceiver **1500** will generally include other circuitry, such as bias circuitry **1546** (which may be distributed throughout the transceiver **1500** in proximity to transistor devices), electrostatic discharge (ESD) protection circuits, testing circuits (not shown), factory programming interfaces (not shown), etc.

In modern transceivers, there are often more than one receiver path Rx and transmitter path Tx, for example, to accommodate multiple frequencies and/or signaling modalities. Further, as should be apparent to one of ordinary skill in the art, some components of the transceiver **1500** may be positioned in a different order (e.g., filters) or omitted. Other components can be (and often are) added, such as (by way of example only) additional filters, impedance matching networks, variable phase shifters/attenuators, power dividers, etc.

As discussed above, the current invention improves IC area savings, manufacturing cost savings, wider impedance matching bandwidth, and flatter gain response of amplifier circuit modules or blocks. As a person of ordinary skill in the art will understand, the system architecture is beneficially impacted by the current invention in critical ways, including better dynamic range, better sensitivity, lower cost, and wider bandwidth. In order to comply with system standards or customer requirements, the current invention is therefore critical to the overall solution shown in FIG. **15**. The current invention therefore specifically defines a system-level embodiment that is creatively enabled by its inclusion in that system.

Methods

Another aspect of the invention includes methods for wideband input impedance matching for an amplifier core having an amplifier input terminal configured to receive a radiofrequency (RF) signal, an amplified-signal terminal, and a degeneration terminal. For example, FIG. **16** is a process flow chart **1600** showing a first method for providing wideband input impedance matching for an amplifier core. The method includes: coupling a first inductor between

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a first terminal and a first node, the first terminal configured to be coupled to the degeneration terminal of the amplifier core (Block **1602**); coupling a second inductor between a second terminal and the first node, the second terminal configured to be coupled to the amplifier input terminal of the amplifier core (Block **1604**); and coupling a third inductor between the first node and a third terminal, the third terminal configured to be coupled to a reference potential (Block **1606**); mutually coupling the first inductor and the second inductor (Block **1608**).

As another example, FIG. **17** is a process flow chart **1700** showing a second method for providing wideband input impedance matching for an amplifier core. The method includes: coupling a first inductor between a first terminal and a first node, the first terminal configured to be coupled to the degeneration terminal of the amplifier core (Block **1702**); coupling a second inductor coupled between a second terminal and a second node, the second terminal configured to be coupled to the amplifier input terminal of the amplifier core (Block **1704**); coupling a third inductor between the first node and a third terminal, the third terminal configured to be coupled to a reference potential (Block **1706**); coupling a fourth inductor between the second node and a fourth terminal, the fourth terminal configured to be coupled to the reference potential (Block **1708**); and mutually coupling the first inductor and the second inductor (Block **1710**).

Additional aspects of the above method may include one or more of the following: wherein the mutual coupling of the first inductor and the second inductor is negative; wherein the mutual coupling of the first inductor and the second inductor is positive; wherein the amplifier core, the first inductor, and the second inductor are co-fabricated on an integrated circuit die; wherein the amplifier core is fabricated on an integrated circuit die, and the first inductor and the second inductor are located external to the integrated circuit die; wherein the first inductor comprises a plurality of parallel inductances; wherein the first inductor and the second inductor comprise mutually coupled segments of an integrated circuit transformer coil; wherein the first inductor comprises a first selectable inductance coupled in series with a second selectable inductance; further including a series inductor coupled between the input terminal of the amplifier core and a radio frequency input terminal of the amplifier core, wherein the second terminal of the wideband input impedance matching network is configured to be coupled to the input terminal of the amplifier core through the series inductor; and/or wherein the second inductor and the first selectable inductance are mutually coupled in a first selectable state, and the second inductor and a series sum of the first and second selectable inductances are mutually coupled in a second selectable state.

Fabrication Technologies & Options

While the example embodiments are LNAs, the inventive circuits and methods may also be applied to other types of amplifiers, such as power amplifiers.

Additional well-known circuit elements that might be included in some applications, such as DC block capacitors, additional impedance matching circuitry, and additional filters, are omitted for clarity. Note also that a circuit component that is characterized as “adjustable” may have its value selected from a number of possible component value settings and fixed during fabrication, when assembled in a circuit module, during factory testing, or in the field (e.g., by burning or “blowing” fusible links), or may have its value be dynamically varied, tuned, or programmatically set, such as

in response to other circuitry (e.g., temperature compensation and/or power control circuitry) or in response to generated or received command signals.

The modes of operation of the inventive LNA circuits may be set by a control circuit (not shown) in known fashion. The control circuit may also connect to the components that are adjustable to select different component values (e.g., capacitance, resistance, inductance) for different gain states, for example, to help input and/or output impedance matching or vary gain versus linearity and/or NF in some modes of operation.

The term “MOSFET”, as used in this disclosure, includes any field effect transistor (FET) having an insulated gate whose voltage determines the conductivity of the transistor, and encompasses insulated gates having a metal or metal-like, insulator, and/or semiconductor structure. The terms “metal” or “metal-like” include at least one electrically conductive material (such as aluminum, copper, or other metal, or highly doped polysilicon, graphene, or other electrical conductor), “insulator” includes at least one insulating material (such as silicon oxide or other dielectric material), and “semiconductor” includes at least one semiconductor material.

As used in this disclosure, the term “radio frequency” (RF) refers to a rate of oscillation in the range of about 3 kHz to about 300 GHz. This term also includes the frequencies used in wireless communication systems. An RF frequency may be the frequency of an electromagnetic wave or of an alternating voltage or current in a circuit.

With respect to the figures referenced in this disclosure, the dimensions for the various elements are not to scale; some dimensions may be greatly exaggerated vertically and/or horizontally for clarity or emphasis. In addition, references to orientations and directions (e.g., “top”, “bottom”, “above”, “below”, “lateral”, “vertical”, “horizontal”, etc.) are relative to the example drawings, and not necessarily absolute orientations or directions.

Various embodiments of the invention can be implemented to meet a wide variety of specifications. Unless otherwise noted above, selection of suitable component values is a matter of design choice. Various embodiments of the invention may be implemented in any suitable integrated circuit (IC) technology (including but not limited to MOSFET structures), or in hybrid or discrete circuit forms. Integrated circuit embodiments may be fabricated using any suitable substrates and processes, including but not limited to standard bulk silicon, high-resistivity bulk CMOS, silicon-on-insulator (SOI), and silicon-on-sapphire (SOS). Unless otherwise noted above, embodiments of the invention may be implemented in other transistor technologies such as bipolar, BiCMOS, LDMOS, BCD, GaAs HBT, GaN HEMT, GaAs pHEMT, and MESFET technologies. However, embodiments of the invention are particularly useful when fabricated using an SOI or SOS based process, or when fabricated with processes having similar characteristics. Fabrication in CMOS using SOI or SOS processes enables circuits with low power consumption, the ability to withstand high power signals during operation due to FET stacking, good linearity, and high frequency operation (i.e., radio frequencies up to and exceeding 300 GHz). Monolithic IC implementation is particularly useful since parasitic capacitances generally can be kept low (or at a minimum, kept uniform across all units, permitting them to be compensated) by careful design.

Voltage levels may be adjusted, and/or voltage and/or logic signal polarities reversed, depending on a particular specification and/or implementing technology (e.g., NMOS,

PMOS, or CMOS, and enhancement mode or depletion mode transistor devices). Component voltage, current, and power handling capabilities may be adapted as needed, for example, by adjusting device sizes, serially “stacking” components (particularly FETs) to withstand greater voltages, and/or using multiple components in parallel to handle greater currents. Additional circuit components may be added to enhance the capabilities of the disclosed circuits and/or to provide additional functionality without significantly altering the functionality of the disclosed circuits.

CONCLUSION

A number of embodiments of the invention have been described. It is to be understood that various modifications may be made without departing from the spirit and scope of the invention. For example, some of the steps described above may be order independent, and thus can be performed in an order different from that described. Further, some of the steps described above may be optional. Various activities described with respect to the methods identified above can be executed in repetitive, serial, and/or parallel fashion.

It is to be understood that the foregoing description is intended to illustrate and not to limit the scope of the invention, which is defined by the scope of the following claims, and that other embodiments are within the scope of the claims. In particular, the scope of the invention includes any and all feasible combinations of one or more of the processes, machines, manufactures, or compositions of matter set forth in the claims below. (Note that the parenthetical labels for claim elements are for ease of referring to such elements, and do not in themselves indicate a particular required ordering or enumeration of elements; further, such labels may be reused in dependent claims as references to additional elements without being regarded as starting a conflicting labeling sequence).

What is claimed is:

1. A wideband coupled input impedance matching network including:
 - (a) a first inductor coupled between a first terminal and a first node, the first terminal configured to be coupled to a degeneration terminal of an amplifier core;
 - (b) a second inductor coupled between a second terminal and the first node, the second terminal configured to be coupled to an input terminal of the amplifier core; and
 - (c) a third inductor coupled between the first node and a third terminal, the third terminal configured to be coupled to a reference potential;
 wherein the first inductor and the second inductor are mutually coupled.
2. The invention of claim 1, wherein the mutual coupling of the first inductor and the second inductor is negative.
3. The invention of claim 1, wherein the mutual coupling of the first inductor and the second inductor is positive.
4. The invention of claim 1, wherein the amplifier core, the first inductor, and the second inductor are co-fabricated on an integrated circuit die.
5. The invention of claim 1, wherein the amplifier core is fabricated on an integrated circuit die, and the first inductor and the second inductor are located external to the integrated circuit die.
6. The invention of claim 1, wherein the first inductor comprises a plurality of parallel inductances.
7. The invention of claim 1, wherein the first inductor and the second inductor comprise mutually coupled segments of an integrated circuit transformer coil.

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8. The invention of claim 1, wherein the first inductor comprises a first selectable inductance coupled in series with a second selectable inductance.

9. The invention of claim 8, wherein the second inductor and the first selectable inductance are mutually coupled in a first selectable state, and the second inductor and a series sum of the first and second selectable inductances are mutually coupled in a second selectable state. 5

10. The invention of claim 1, further including a series inductor coupled between the input terminal of the amplifier core and a radio frequency input terminal of the amplifier core, wherein the second terminal of the wideband input impedance matching network is configured to be coupled to the input terminal of the amplifier core through the series inductor. 10 15

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